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[647] ON THE QUARTIC SURFACES REPRESENTED BY THE EQUATION,
SYMMETRICAL DETERMINANT = 0 (CONTINUED).

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 46–52;
continued from preceding pages.]

(Page 51.) I, in fact, take the terms $(a, \dots)$ of the determinant to be homogeneous functions of (x, y, z, w) of the degrees

$$\begin{array}{cccc} 0, & 1, & 1, & 0 \\ 1, & 2, & 2, & 1 \\ 1, & 2, & 2, & 1 \\ 0, & 1, & 1, & 0 \end{array}$$

respectively, viz. a, d, l are constants, g, h, m, n linear functions, and b, c, f quadric functions of the coordinates; $\nabla = 0$ still represents a quartic surface; and it appears by a general formula that the number of nodes is = 8. But we can easily show this directly; and further, that the 8 nodes are the intersections of three quadric surfaces; or say that the quartic surface is octadic. For denoting as before the first minors by $A, \dots$, then B, C, F are each of them a quadric function of the coordinates, viz. we have

$$\begin{aligned} B &= d(ac - g^2) - cl^2 - an^2 + 2gln, \\ C &= d(ab - h^2) - am^2 - bl^2 + 2hlm, \\ F &= d(gh - af) + l^2f + mna - nlh - lmg, \end{aligned}$$

and we have identically

$$BC - F^2 = (ad - l^2)\nabla,$$

so that throwing out the constant factor $ad - l^2$, the equation of the surface is

$$BC - F^2 = 0,$$

and it has 8 nodes, the intersections of the three quadric surfaces $B = 0, C = 0, F = 0$. By equating to zero any other minor of the determinant ∇ , we have a surface passing through the 8 nodes; we have for instance the quartic surface

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix}^2 = 0.$$

Suppose now (and in all that follows) that, the degrees being as already mentioned, we further assume that b, c, f are quadric functions of the form $(x, y)^2$, g, h linear functions of the form (x, y) ; then since each term of ∇ contains either

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix},$$

or one of its first minors, it is clear that the line $(x = 0, y = 0)$ is a double line on the surface. But in the present case there is not any diminution in the number of the nodes; in fact, writing $x = 0, y = 0$, and therefore b, c, f, g, h each $= 0$ (but not $a = 0$), the minors B, C, F none of them vanish; that is, the line $x = 0, y = 0$ is not a line on any one of the quadric surfaces, and the quadric surfaces intersect as before in an octad of points.

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(Page 52.) The equation $\nabla = 0$ thus represents a quartic surface having a double line, and also 8 nodes forming an octad.

We may without loss of generality write $d = 0$; in fact, the determinant is unaltered if we add to the fourth column θ times the first column, and then to the fourth line θ times the first line; the determinant is thus of the original form, but in place of d it has $d + 2\theta l + \theta^2 a$, which by properly determining θ can be made $= 0$. And then changing the original l, m, n , the equation is

$$\nabla = \begin{vmatrix} a & h & g & l \\ h & b & f & m \\ g & f & c & n \\ l & m & n & 0 \end{vmatrix} = 0.$$

Or, writing for shortness,

$$K = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix},$$

and denoting the minors hereof by (a, b, c, f, g, h) , then the equation is

$$\nabla = (a, b, c, f, g, h \mid l, m, n)^2 = 0,$$

where the degree of K is 4, and the degrees of a, b, c, f, g, h are 4, 2, 2, 2, 3, 3 respectively, those of l, m, n being 0, 1, 1 respectively.

The nodes are, as before, the intersections of the quadric surfaces $B = 0, C = 0, F = 0$, viz. (d being now $= 0$) the values are

$$\begin{aligned} -B &= cl^2 - 2gln + an^2, \\ -C &= bl^2 - 2hlm + am^2, \\ F &= fl^2 - glm - hln + amn. \end{aligned}$$

But, according to a previous remark, the nodes lie also on the quartic surface $K^2 = 0$; viz. this is a set of four planes intersecting in the line $x = 0, y = 0$.

Now, in general, any plane through the line $x = 0, y = 0$ meets the surface in this line twice and in a conic; if the plane is $y = \theta x$, we have

$$(a, b, c, f, g, h \mid x, y)^2 = (a, b'x^2, c'x^2, f'x^2, g'x, h'x),$$

where a, b', c', f', g', h' are functions of θ of the degrees (0, 2, 2, 2, 1, 1) respectively; and thence also

$$(a, b, c, f, g, h) = (a', b'x^2, c'x^2, f'x^2, g'x^3, h'x^3),$$

where $\mathbf{a}', \mathbf{b}', \mathbf{c}', \mathbf{f}', \mathbf{g}', \mathbf{h}'$ are functions of θ of the degrees 4, 2, 2, 2, 3, 3 respectively; the equation of the surface thus becomes $(\mathbf{a}', \mathbf{b}', \mathbf{c}', \mathbf{f}', \mathbf{g}', \mathbf{h}' \mid lx, m, n)^2 = 0$; viz. this is a quadric equation which, combined with the equation $y - \theta x = 0$, determines the

(Page 53.) conic in question. But for each of the planes $K = 0$, we have $(\mathbf{a}', \mathbf{b}', \mathbf{c}', \mathbf{f}', \mathbf{g}', \mathbf{h}' \mid lx, m, n)^2$, a perfect square, or the conic a two-fold line; we have thus the 8 nodes lying in pairs on four lines, say the four “rays,” in the four planes $K = 0$ respectively; each of these rays meets the double line $x = 0, y = 0$ in a point; and we have thus on the double line 4 points, which are in fact pinch-points of the surface (as to this presently). It has just been stated that for the plane passing through the nodal line and a ray, the conic is a two-fold line (the ray twice) containing upon it a pair of nodes; more properly, the conic is the point-pair composed of the two nodes.

We can find through the nodes four different plane-pairs; in fact, forming the equation

$$B - 2\lambda F + C\lambda^2 = 0,$$

this is

$$l^2(c + 2\lambda f + \lambda^2 b) - 2l(g + \lambda h)(n + \lambda m) + a(n + \lambda m)^2 = 0;$$

or, as this may also be written,

$$[a(n + \lambda m) - l(g + \lambda h)]^2 + l^2 a(b - 2\lambda f + \lambda^2 c) = 0,$$

where b, c, f and therefore also $b - 2\lambda f + \lambda^2 c$ are of the form $(x, y)^2$; say that we have

$$b - 2\lambda f + \lambda^2 c = (p, q, r \mid x, y)^2,$$

where p, q, r are of course quadric functions of λ ; determining λ by the quartic equation $pr - q^2 = 0$, we have $b - 2\lambda f + \lambda^2 c$ a perfect square, $= (\alpha x + \beta y)^2$ suppose; and we have thus the plane-pair

$$[a(n + \lambda m) - l(g + \lambda h)]^2 - l^2 a(\alpha x + \beta y)^2 = 0$$

containing the eight nodes; viz. there are four such plane-pairs. The two planes of a plane-pair intersect in a line called an “axis”; that is, we have four axes each meeting the nodal line; and we have thus also through the nodal line and the four axes respectively four planes, which are “pinch-planes” of the quartic surface (as to this presently).

It has just been seen that the equation $B - 2\lambda F + C\lambda^2 = 0$ (where λ is arbitrary) is expressible in the form

$$[a(n + \lambda m) - l(g + \lambda h)]^2 + l^2 a(p, q, r \mid x, y)^2 = 0,$$

viz. this is the equation of a quadric cone having its vertex on the nodal line at the point $x = 0, y = 0, an - lg + \lambda(am - lh) = 0$; this is, in fact, a cone touching the surface, as at once appears by writing the equation of the cone in the form

$$\frac{1}{C}\{BC - F^2 + (\lambda C - F)^2\} = 0,$$

that is,

$$\frac{1}{C}[-l^2 a \nabla + (\lambda C - F)^2] = 0;$$

we thus see that, taking for vertex any point whatever on the nodal line, there is a circumscribed quadric cone.

(Page 54.) For each of the above-mentioned four values of λ , the quadric cone breaks up into a plane-pair; each plane of the plane-pair is thus a “trope” or plane touching the surface along a conic; viz. this is the conic passing through the intersection of the plane (or say of an axis) with the nodal line and through four nodes of the surface. We have thus 8 tropes, intersecting in pairs in the four axes (and the intersection of each axis with the nodal line being a pinch-point). Moreover, joining the nodes in pairs, we have four rays, each meeting the nodal line, the plane through it and the nodal line being a pinch-plane; this is illustrated in the figure.

[Figure: schematic showing nodal line, axis, pinch-point and pinch-planes.]

As to the pinch-planes and pinch-points, remark first that a plane through the nodal line is in general a bitangent plane, its two points of contact being the points where the conic in such plane meets the nodal line. When the two points of contact come to coincide, the plane is a pinch-plane; viz. this happens when the plane passes through a ray, the conic being then the ray twice repeated. And secondly, at a point on the nodal line there are in general two tangent planes, viz. these are the tangent planes to the quadric cone belonging to such point; when the two tangent-planes come to coincide the point is a pinch-point, and this happens when the point is the intersection of the nodal line with an axis, for then (the quadric cone breaking up into the two tropes through the axis) the two tangent planes become the plane through the axis taken twice.

Each section through the nodal line is a conic, and the polar of the nodal line in regard to this conic is a point; the locus of this point (for different sections through the nodal line) is a right line which may be called simply the “polar.” To prove this, considering the section by the plane $y = \theta x$, we have to find the pole of the line $x = 0$ in regard to the conic

$$(\mathfrak{a}', \mathfrak{b}', \mathfrak{c}', \mathfrak{f}', \mathfrak{g}', \mathfrak{h}' \mid lx, m, n)^2 = 0;$$

(Page 55.) this is $lx : m : n = \mathfrak{a}' : \mathfrak{h}' : \mathfrak{g}'$, viz. if $\mathfrak{g} = g_0x + g_1y$, $\mathfrak{h} = h_0x + h_1y$, this is

$$lx : m : n = a : g_0 + g_1\theta : h_0 + h_1\theta,$$

or joining hereto the equation $y = \theta x$, we have

$$lx : ly : m : n = a : a\theta : g_0 + g_1\theta : h_0 + h_1\theta,$$

where l, a, g_0, g_1, h_0, h_1 are constants; m, n are linear functions of the coordinates (x, y, z, w) . The equations represent, it is clear, a right line which is the polar in question; and they may be written

$$\frac{lx}{a} = \frac{h_1m - g_1n}{h_0g_1 - h_1g_0}, \quad \frac{ly}{a} = \frac{h_0m - g_0n}{h_1g_0 - h_0g_1}.$$

When the plane passes through a ray, the conic becomes, as was stated, the point-pair composed of the two nodes in such ray; the harmonic in regard to these two points of the intersection of the ray with the nodal line is thus a point on the polar: that is, the polar meets the ray; and the two nodes are situate harmonically in regard to the intersections of the ray with the nodal line and the polar respectively.

The polar may be arrived at in a different manner, viz. if instead of a plane through the nodal line we consider a point on the nodal line, this is the vertex of a circumscribed quadric cone; and taking the polar plane of the nodal line in regard to this cone, then considering the point as variable, the different polar planes all pass through a line which is the polar in question. And hence, taking for the point the intersection of the nodal line with an axis, it appears that the axis meets the polar; and, moreover, that the two tropes through the axis are harmonics in regard to the planes through the axis, and the polar and nodal line respectively.

Collecting the foregoing results, we have a quartic surface as follows:

We have two lines, a nodal line and a polar; meeting each of these, four lines called “rays” and four other lines called “axes.” On each ray, harmonically in regard to its intersections with the nodal line and the polar, two nodes of the surface (in all 8 nodes); through each axis, harmonically in regard to the planes through it and the nodal line and the axis respectively, two tropes of the surface (in all 8 tropes). In each trope (or, what is the same thing, in its conic of contact) are 4 nodes; through each node (or, what is the same thing, touching its tangential quadricone) are 4 tropes; the relation of the nodes and tropes may be thus represented, viz. taking the pairs of

nodes to be 1, 2; 3, 4; 5, 6; 7, 8; and those of tropes to be I, II; III, IV; V, VI; VII, VIII; then we have

(Page 56.)

	I	II	III	IV	V	VI	VII	VIII
1	•				•		•	
2		•				•		•
3			•		•			•
4				•		•	•	
5	•		•					•
6		•		•			•	
7	•			•		•		
8		•	•		•			

viz. reading horizontally or vertically, the dots show the tropes through each node, or the nodes in each trope.

The plane through any ray and the nodal line is a pinch-plane of the surface, its point of contact being the intersection of the ray with the nodal line; and the intersection of each axis with the nodal line is a pinch-point of the surface, the tangent plane being the plane through the axis and the nodal line; the surface has thus 4 pinch-planes and 4 pinch-points.

648.

ALGEBRAICAL THEOREM.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), p. 53.]

(Page 57.) I WISH to put on record the following theorem, given by me as a Senate-House Problem, January, 1851.

If $\{\alpha + \beta + \gamma + \dots\}^p$ denote the expansion of $(\alpha + \beta + \gamma + \dots)^p$, retaining those terms $N\alpha^a\beta^b\gamma^c\dots$ only in which

$$b + c + d + \dots > p - 1, \quad c + d + \dots > p - 2, \quad \&c., \quad \&c.,$$

then

$$\begin{aligned} x^n &= (x + a)^n - n\{\alpha\}^1(x + a + \beta)^{n-1} + \frac{1}{2}n(n-1)\{\alpha + \beta\}^2(x + a + \beta + \gamma)^{n-2} \\ &\quad - \frac{1}{6}n(n-1)(n-2)\{\alpha + \beta + \gamma\}^3(x + a + \beta + \gamma + \delta)^{n-3} + \&c. \end{aligned}$$

The theorem, in a somewhat different and imperfectly stated form, is given, Burg, *Crelle*, t. i. (1826), p. 368, as a generalisation of Abel's theorem,

$$\begin{aligned} (x + a)^n &= x^n + na(x + \beta)^{n-1} + \frac{1}{2}n(n-1)a(a-2\beta)(x + 2\beta)^{n-2} \\ &\quad + \frac{1}{6}n(n-1)(n-2)a(a-3\beta)^2(x + 3\beta)^{n-3} + \&c. \end{aligned}$$

649.

**ADDITION TO MR GLAISHER'S NOTE ON SYLVESTER'S PAPER,
"DEVELOPMENT OF AN IDEA OF EISENSTEIN."**

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 83, 84.]

(Page 58.) THE formula (11) [in the Note], under a slightly different form, is demonstrated by me in an addition [263] to Sir J. F. W. Herschel's paper "On the formulae investigated by Dr Brinkley, &c.," *Phil. Trans.*, t. CL., 1860, pp. 321–323. The demonstration is in effect as follows: let u denote a series of the form $1 + bx + cx^2 + dx^3 + \dots$, and let u^i (where i is positive or negative, integer or fractional) denote the development of the i -th power of u , continued up to the term which involves x^n , the terms involving higher powers of x being rejected; u^0, u^1, u^2 , and generally u^s will denote in like manner the developments of these powers up to the terms involving x^n , or, what is the same thing, they will be the values of u^i corresponding to $i = 0, 1, 2, \dots, s$. By the formula

$$u^i = 1 + \frac{i}{1}(u-1) + \frac{i(i-1)}{1 \cdot 2}(u-1)^2 + \dots \text{ as far as the term involving } (u-1)^n, \text{ } u^i \text{ is a rational}$$

and integral function of i of the degree n , and can therefore be expressed in terms of the values $u^0, u^1, u^2, \dots, u^n$ which correspond to $i = 0, 1, 2, \dots, n$. Let s have any one of the last-mentioned values, then the expression

$$\frac{i \cdot i - 1 \cdot i - 2 \dots i - n}{i - s} \cdot \frac{1}{s \cdot s - 1 \dots 2 \cdot 1 \cdot -1 \cdot -2 \dots - (n - s)},$$

which as regards i is a rational and integral function of the degree n (the factor $i - s$ which occurs in the numerator and denominator being of course omitted), vanishes for each of the values $i = 0, 1, 2, \dots, n$, except only for the value $i = s$, in which case it becomes equal to unity. The required formula is thus seen to be

$$u^i = \Sigma \left\{ \frac{i \cdot i - 1 \cdot i - 2 \dots i - n}{i - s} \cdot \frac{1}{s \cdot s - 1 \dots 2 \cdot 1 \cdot -1 \cdot -2 \dots - (n - s)} u^s \right\},$$

(Page 59.) where the summation extends to the several values $s = 0, 1, 2, \dots, n$; or, what is the same thing, it is

$$u^i = \Sigma \left\{ \frac{i \cdot i - 1 \cdot i - 2 \dots i - n}{i - s} \cdot \frac{(-)^{n-s} 1}{1 \cdot 2 \dots s \cdot 1 \cdot 2 \dots n - s} u^s \right\},$$

or, changing the sign of i , it is

$$u^{-i} = \Sigma \left\{ \frac{i \cdot i + 1 \cdot i + 2 \dots i + n}{i + s} \cdot \frac{(-)^s 1}{1 \cdot 2 \dots s \cdot 1 \cdot 2 \dots n - s} u^s \right\},$$

where, as before, s has the values $0, 1, 2, \dots, n$ successively. Or, what is the same thing, we have

$$C_{-i,n} = \Sigma \left\{ \frac{i \cdot i + 1 \cdot i + 2 \dots i + n}{i + s} \cdot \frac{(-)^s 1}{1 \cdot 2 \dots s \cdot 1 \cdot 2 \dots n - s} C_{s,n} \right\},$$

where the term corresponding to $s = 0$, as containing the factor $C_{0,n}$ vanishes except in the case $n = 0$ (for which it is = 1); and omitting this evanescent term, this is in fact the formula (11).

650.

ON A QUARTIC SURFACE WITH TWELVE NODES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 103–106.]

(Page 60.) WRITE for shortness

$$\begin{aligned} a &= \beta - \gamma, & f &= \alpha - \delta, & af &= p, \\ b &= \gamma - \alpha, & g &= \beta - \delta, & bg &= q, \\ c &= \alpha - \beta, & h &= \gamma - \delta, & ch &= r; \end{aligned}$$

then, θ being a variable parameter, the surface in question is the envelope of the quadric surface

$$(\alpha + \theta)^2 aghX^2 + (\beta + \theta)^2 bhfY^2 + (\gamma + \theta)^2 cfgZ^2 + (\delta + \theta)^2 abcW^2 = 0;$$

viz. this is

$$\Sigma \theta^2 aghX^2 \cdot \Sigma aghX^2 - \Sigma a aghX^2 = 0.$$

There are no terms in X^4 , &c.; the coefficient of Y^2Z^2 is

$$\gamma^2 cfg \cdot bfh + \beta^2 bfh \cdot cfg - 2\beta\gamma bfh \cdot cfg,$$

which is

$$= bcfgh(\beta - \gamma)^2, = a^2bcf^2gh, = abcfgh \cdot p.$$

Hence the whole equation divides by $abcfgh$, and throwing out this factor, the result is

$$p(Y^2Z^2 + X^2W^2) + q(Z^2X^2 + Y^2W^2) + r(X^2Y^2 + Z^2W^2) = 0,$$

or, observing that $p + q + r = 0$, this may also be written

$$p(YZ + XW)^2 + q(ZX + YW)^2 + r(XY + ZW)^2 = 0,$$

(Page 61.) and also

$$p(YZ - XW)^2 + q(ZX - YW)^2 + r(XY - ZW)^2 = 0.$$

The more general equation

$$(p, q, r, l, m, n \mid YZ + XW, ZX + YW, XY + ZW)^2 = 0$$

represents a quartic surface (octadic) having the 8 nodes

$$\begin{aligned} (1, 0, 0, 0), & \quad (\bar{1}, 1, 1, 1), \\ (0, 1, 0, 0), & \quad (1, \bar{1}, 1, 1), \\ (0, 0, 1, 0), & \quad (1, 1, \bar{1}, 1), \\ (0, 0, 0, 1), & \quad (1, 1, 1, \bar{1}). \end{aligned}$$

We have

$d_x U =$	$d_y U =$
$p \cdot XW^2 + YZW$	$p \cdot YZ^2 + XZW$
$q \cdot YW^2 + YZW$	$q \cdot YW^2 + XZW$
$r \cdot ZW^2 + YZW$	$r \cdot YX^2 + XZW$
$l \cdot 2XYZ + W(Y^2 + Z^2)$	$l \cdot 2XYW + Z(W^2 + X^2)$
$m \cdot 2XYW + Z(W^2 + Y^2)$	$m \cdot 2YZX + W(Z^2 + X^2)$
$n \cdot 2XZW + Y(W^2 + Z^2)$	$n \cdot 2YZW + X(W^2 + Z^2),$

$d_z U =$	$d_w U =$
$p \cdot Y^2 Z + XYW$	$p \cdot X^2 W + XYZ$
$q \cdot X^2 Z + XYW$	$q \cdot Y^2 W + XYZ$
$r \cdot W^2 Z + XYW$	$r \cdot Z^2 W + XYZ$
$l \cdot 2ZXW + Y(W^2 + X^2)$	$l \cdot 2WYZ + X(Y^2 + Z^2)$
$m \cdot 2YZW + X(W^2 + Y^2)$	$m \cdot 2WZX + Y(Z^2 + X^2)$
$n \cdot 2ZXY + W(X^2 + Y^2)$	$n \cdot 2WXY + Z(X^2 + Y^2)$

Hence there will be a node

$$\begin{aligned}
 1, 1, 1, 1, \quad & \dots \quad p + q + r + 2l + 2m + 2n = 0, \\
 \bar{1}, 1, 1, 1, \quad & \dots \quad p + q + r - 2l + 2m - 2n = 0, \\
 1, \bar{1}, 1, 1, \quad & \dots \quad p + q + r + 2l - 2m + 2n = 0, \\
 1, 1, \bar{1}, 1, \quad & \dots \quad p + q + r - 2l - 2m + 2n = 0;
 \end{aligned}$$

or say there will be

$$\begin{aligned}
 1 \text{ of these nodes } & p + q + r + 2l + 2m + 2n = 0, \\
 2 & p + q + r + 2l = 0, \quad m + n = 0, \\
 3 & p + q + r = 2l = -2m = -2n, \\
 4 & p + q + r = 0, \quad l = 0, \quad m = 0, \quad n = 0;
 \end{aligned}$$

(Page 62.) viz. the surface having the 12 nodes is the original surface

$$p(YZ + XW)^2 + q(ZX + YW)^2 + r(XY + ZW)^2,$$

where

$$p + q + r = 0.$$

The Jacobian of the quadrics

$$YZ + XW = 0, \quad ZX + YW = 0, \quad XY + ZW = 0,$$

is

$$\begin{vmatrix} W & Z & Y & X \\ Z & W & X & Y \\ Y & X & W & Z \end{vmatrix} = 0;$$

viz. the equations are

$$\begin{aligned}
 X^3 - X(Y^2 + Z^2 + W^2) + 2YZW &= 0, \\
 Y^3 - Y(Z^2 + X^2 + W^2) + 2ZXW &= 0, \\
 Z^3 - Z(X^2 + Y^2 + W^2) + 2XYW &= 0, \\
 W^3 - W(X^2 + Y^2 + Z^2) + 2XYZ &= 0,
 \end{aligned}$$

each of which is satisfied in virtue of any one of the pairs of equations

$$\begin{aligned}
 (Y - Z = 0, \quad X - W = 0) \quad & (Y + Z = 0, \quad X + W = 0), \\
 (Z - X = 0, \quad Y - W = 0) \quad & (Z + X = 0, \quad Y + W = 0), \\
 (X - Y = 0, \quad Z - W = 0) \quad & (X + Y = 0, \quad Z + W = 0).
 \end{aligned}$$

so that the Jacobian curve is, in fact, the six lines represented by these equations.

Any two of the three tetrads form an octad, the 8 points of intersection of three quadric surfaces: a figure representing the relation of the 12 points to each other may be constructed without difficulty.

Each tetrad is a sibi-conjugate tetrad *quoad* the quadric $X^2 + Y^2 + Z^2 + W^2 = 0$. The three tetrads are not on the same quadric surface.

651.

ON A SPECIAL SURFACE OF MINIMUM AREA.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 190–196.]

(Page 63.) A VERY remarkable form of the surface of minimum area was obtained by Prof. Schwarz in his memoir “Bestimmung einer speciellen Minimal-fläche,” Berlin, 1871, [*Ges. Werke*, t. I., pp. 6–125], crowned by the Academy of Sciences at Berlin. The equation of the surface is

$$1 + \mu\nu + \nu\lambda + \lambda\mu = 0,$$

where λ, μ, ν are functions of x, y, z respectively, viz.

$$x = - \int \frac{d\theta}{\sqrt{(\frac{1}{4}\theta^4 + \frac{1}{2}\theta^2 + \frac{1}{4})}},$$

and y, z are the same functions of μ, ν respectively. A direct verification of the theorem that this is a surface of minimum area, satisfying, that is, the differential equation

$$r(1 + q^2) - 2pqs + t(1 + p^2) = 0,$$

is given in the memoir; but the investigation may be conducted in quite a different manner, so as to be at once symmetrical and somewhat more general, viz. we may enquire whether there exists a surface of minimum area

$$1 + \mu\nu + \nu\lambda + \lambda\mu = 0,$$

where the determining equations are

$$\begin{aligned}\lambda'^2 &= a\lambda^4 + b\lambda^2 + c, \\ \mu'^2 &= a\mu^4 + b\mu^2 + c, \\ \nu'^2 &= a\nu^4 + b\nu^2 + c,\end{aligned}$$

(Page 64.) ($\lambda' = \frac{d\lambda}{dx}$, &c.). I find that the coefficients a, b, c must satisfy four homogeneous quadric equations, which, in fact, admit of simultaneous solution, and that in three distinct ways; viz. assuming $a = 1$, the solutions are

$$\begin{aligned}a = 1, \quad b = \frac{1}{2}, \quad c = 1, \\ a = 1, \quad b = -2, \quad c = 1, \\ a = 1, \quad b = -\frac{3}{2}, \quad c = -\frac{1}{2};\end{aligned}$$

that is,

$$\lambda'^2 = \lambda^4 + \frac{1}{2}\lambda^2 + 1 \quad \{= \frac{1}{4}(\frac{4}{3}\lambda^4 + \frac{2}{3}\lambda^2 + \frac{4}{3})\},$$

which gives Schwarz's surface;

$$\lambda'^2 = \lambda^4 - 2\lambda^2 + 1 \text{ or } \lambda' = \pm(\lambda^2 - 1),$$

which, it is easy to see, gives only $x + y + z = \text{const.}$; and

$$\lambda'^2 = \lambda^4 - \frac{3}{2}\lambda^2 - \frac{1}{2}, = (\lambda^2 - 1)(\lambda^2 + \frac{1}{2}),$$

which is a surface similar in its nature to Schwarz's surface.

The investigation is as follows: the condition to be satisfied by a surface of minimum area $U = 0$ is

$$(\mathbf{a} + \mathbf{b} + \mathbf{c})(X^2 + Y^2 + Z^2) - (\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h} \mid X, Y, Z)^2 = 0,$$

where (X, Y, Z) are the first derived coefficients and $(\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h})$ the second derived coefficients of U in regard to the coordinates. Considering U as a function of λ, μ, ν , which are functions of x, y, z respectively, and writing (L, M, N) and $(\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h})$ for the first and second derived functions of U in regard to λ, μ, ν , also λ', λ'' for the first and second derived functions of λ in regard to x , and so for μ, μ'' and ν, ν'' we have:

$$(X, Y, Z) = (L\lambda', M\mu', N\nu'),$$

$$(\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h}) = (\mathbf{a}\lambda'^2 + L\lambda'', \mathbf{b}\mu'^2 + M\mu'', \mathbf{c}\nu'^2 + N\nu'', \mathbf{f}\mu'\nu', \mathbf{g}\nu'\lambda', \mathbf{h}\lambda'\mu'),$$

and for the particular surface $U = 1 + \mu\nu + \nu\lambda + \lambda\mu = 0$, the values are

$$(L, M, N, \mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h}) = (\mu + \nu, \nu + \lambda, \lambda + \mu, 0, 0, 0, 1, 1, 1).$$

Hence the condition is found to be

$$\begin{aligned} & 2\mu'^2\nu'^2(\lambda + \mu)(\lambda + \nu) \\ & + 2\nu'^2\lambda'^2(\mu + \nu)(\mu + \lambda) \\ & + 2\lambda'^2\mu'^2(\nu + \lambda)(\nu + \mu) \\ & - \lambda''(\mu + \nu)\{(\lambda + \nu)^2\mu'^2 + (\lambda + \mu)^2\nu'^2\} \\ & - \mu''(\nu + \lambda)\{(\mu + \lambda)^2\nu'^2 + (\mu + \nu)^2\lambda'^2\} \\ & - \nu''(\lambda + \mu)\{(\nu + \mu)^2\lambda'^2 + (\nu + \lambda)^2\mu'^2\} = 0, \end{aligned}$$

(Page 65.) or say this is

$$\Sigma 2\mu'^2\nu'^2(\lambda + \mu)(\lambda + \nu) - \Sigma \lambda''(\mu + \nu)\{(\lambda + \nu)^2\mu'^2 + (\lambda + \mu)^2\nu'^2\} = 0.$$

We have to write in this equation $\lambda'^2 = a\lambda^4 + b\lambda^2 + c$, and therefore $\lambda'' = 2a\lambda^3 + b\lambda$, &c.; the left-hand side, call it Ω , is a symmetrical function of λ, μ, ν , and is consequently expressible as a rational function of

$$\begin{aligned} p, &= \lambda + \mu + \nu, \\ q, &= \mu\nu + \nu\lambda + \lambda\mu, \\ r, &= \lambda\mu\nu. \end{aligned}$$

We ought to have $\Omega = 0$, not identically, but in virtue of the equation $1 + q = 0$, that is, Ω should divide by $1 + q$; or, what is the same thing, Ω should vanish on writing therein $q = -1$.

To effect the reduction as easily as possible, observe that we have $(\lambda + \mu)(\lambda + \nu) = \lambda^2 + q$; and therefore

$$\Sigma \mu'^2\nu'^2(\lambda + \mu)(\lambda + \nu) = \Sigma \lambda^2\mu'^2\nu'^2 + q\Sigma \mu'^2\nu'^2.$$

Similarly, in the second term,

$$(\mu + \nu)(\lambda + \nu)^2 = (\nu + \lambda)(\nu^2 + q) \text{ and } (\mu + \nu)(\lambda + \mu)^2 = (\mu + \lambda)(\mu^2 + q).$$

The complete value of Ω thus is

$$\Omega = 2(Aq + B) - [(C + D)q + E + F],$$

where

$$\begin{aligned} A &= \Sigma \lambda^2 \mu'^2 \nu'^2, & B &= \Sigma \mu'^2 \nu'^2, \\ C &= \Sigma \lambda \lambda'' (\nu \mu'^2 + \mu \nu'^2), & D &= \Sigma \lambda'' (\nu \mu'^2 + \mu \nu'^2), \\ E &= \Sigma \lambda \lambda'' (\mu'^2 + \nu'^2), & F &= \Sigma \lambda'' (\nu \mu'^2 + \mu \nu'^2). \end{aligned}$$

We find without difficulty

$$\begin{aligned} A &= a^2 (q^4 - 4q^2 pr + 2r^2 + 2p^2 r^2) \\ &\quad + ab (-2q^3 + qp^2 + 4qr^2 - 3r^2 - 2p^2 r) \\ &\quad + ac (4q^2 - 3q^2 p^2 + 8pr + 2p^2) \\ &\quad + b^2 (q^2 - 2pr) \\ &\quad + bc (-4q + 2p^2) \\ &\quad + c^2 (3), \end{aligned}$$

$$\begin{aligned} B &= a^2 (q^2 r^2 + 2pr^3) \\ &\quad + ab (-4qr^2 + 2p^2 r^2) \\ &\quad + ac (-2q^3 + q^2 p^2 + 4qpr - 3r^2 - 2p^2 r) \\ &\quad + b^2 (3r^2) \\ &\quad + bc (2q^2 - 4pr) \\ &\quad + c^2 (-2q + 2p^2), \end{aligned}$$

(Page 66.)

$$\begin{aligned} C &= a^2 (4q^2 - 16q^2 pr + 16pr^2 + 8p^2 r^2) \\ &\quad + ab (-6q^3 + 3q^2 p^2 + 12qpr - 9r^2 - 6p^2 r) \\ &\quad + ac (8q^2 - 16qp^2 + 16pr + 4p^4) \\ &\quad + b^2 (2q^2 - 4pr) \\ &\quad + bc (-4q + 2p^2), \end{aligned}$$

$$\begin{aligned} D &= a^2 (4q^2 pr - 2qr^2 - 4p^2 r^2) \\ &\quad + ab (-4qpr + 2p^2 r) \\ &\quad + ac (-4q^2 + 2qp^2 - 2pr) \\ &\quad + b^2 (2pr) \\ &\quad + bc (2q), \end{aligned}$$

$$\begin{aligned} E &= a^2 (4q^2 r - 4pr^2) \\ &\quad + ab (-12qr^2 + 6p^2 r^2) \\ &\quad + ac (-4q^2 + 2qp^2 + 8qpr - 6r^2 - 4p^2 r) \\ &\quad + bc (2q - 4pr), \end{aligned}$$

$$\begin{aligned} F &= a^2 (4pr^2) \\ &\quad + ab () \\ &\quad + ac (4q^2 - 12qpr + 12p^2 r) \\ &\quad + b^2 (qpr - 3r^2) \\ &\quad + bc (-2q^2 + qp^2 - pr), \end{aligned}$$

where in each line the terms are arranged according to their order in p, r .

Substituting, we find

$$\begin{aligned}\Omega = & a^2 (-2q^3 + 6q^2pr - 8qr^2) \\ & + ab (2q^2 - q^2pr + qr^2 - 4r^2) \\ & + ac [-2q^3p^2 + 14qpr - 12r^2] \\ & + b^2 (-3qpr + 3r^2) \\ & + bc (-2q^2 + qp^2 - 3pr) \\ & + c^2 (2q + 2p^2);\end{aligned}$$

viz. writing $q = -1$, this is

$$\begin{aligned}\Omega = & a^2 (2 - 6pr - 8r^2) \\ & + ab (2 + p^2 - pr - 4r^2) \\ & + ac (-2p^2 - 14pr - 12r^2) \\ & + b^2 (3pr + 3r^2) \\ & + bc (-2 - p^2 - 3pr) \\ & + c^2 (-2 - 2p^2);\end{aligned}$$

(Page 67.) or, what is the same thing, it is

$$\begin{aligned}& (2a^2 + 2ab - 2bc - 2c^2) \\ & + p^2 (ab - 2ac - bc + 2c^2) \\ & + pr (-6a^2 - ab - 14ac + 3b^2 - 3bc) \\ & + r^2 (-8a^2 - 4ab - 12ac + 3b^2)\end{aligned}$$

so that, writing for convenience $a = 1$, the equations to be satisfied are

$$\begin{aligned}2 - 2c^2 + 2(1 - c)b &= 0, \\ -2c + 2c^2 + (1 - c)b &= 0, \\ -6 - 14c + 3b^2 - (1 + 3c)b &= 0, \\ -8 - 12c + 3b^2 - 4b &= 0.\end{aligned}$$

The first and second are $(1 - c)(2 + 2c + 2b) = 0$ and $(1 - c)(-2c + b) = 0$; viz. they give $c = 1$, or else $b = -\frac{1}{2}$, $c = \frac{1}{2}$. In the former case, the third and fourth equations each become $3b^2 - 4b - 20 = 0$, that is $(3b - 10)(b + 2) = 0$; in the latter case, they are satisfied identically; hence we have for a, b, c the three systems of values mentioned at the beginning.

This completes the investigation; but it is interesting to find the values assumed by the other factor of Ω on substituting therein for a, b, c the foregoing several systems of values. We have

in general

$$\begin{aligned}
 \Omega &= -2a^2q^3 + 2abq^2 - 2bcq^2 + 2c^2q \\
 &\quad + p^2(-abq^2 - 2acq^2 + bcq + 2c^2) \\
 &\quad + pr[6a^2q^2 - abq + (14ac - 3b^2)q - 3bc] \\
 &\quad + r^2[-8a^2q + 4abq - 12ac + 3b^2] \\
 &= -2a^2(q^3 - q^2) + 2ab(q^2 - 1) - 2bc(q^2 - 1) + 2c^2(q + 1) \\
 &\quad + p^2[-ab(q^2 + 1) - 2ac(q^2 - 1) + bc(q + 1)] \\
 &\quad + pr[6a^2(q^2 + q + 1) - ab(q - 1) + (14ac - 3b^2)(q + 1)] \\
 &\quad + r^2\{-8a^2(q - 1) + 4ab\} \\
 &= (q + 1)\{-2a^2(q^2 - q + q^2 - q + 1) + 2ab(q^2 - q + q - 1) - 2bc(q - 1) + 2c^2 \\
 &\quad + p^2[-ab(q^2 - q - 1) - 2ac(q - 1) + bc] \\
 &\quad + pr[6a^2(q^2 + q + 1) - ab(q - 1) + (14ac - 3b^2)] \\
 &\quad + r^2\{-8a^2(q - 1) + 4ab\}\}.
 \end{aligned}$$

Hence writing, first, $a = c = 1$, $b = -\frac{1}{2}$, we obtain, after some reductions,

$$\Omega = (q + 1)\{-2q(q - 1)(q^2 - \frac{1}{4}q + 1) + p^2(q - 1)(-\frac{1}{2}q - 2) + pr(6q^2 - \frac{1}{4}q - 10) + r^2(-\frac{4}{3} + \frac{1}{2})\};$$

secondly, writing $a = c = 1$, $b = -2$, we obtain

$$\Omega = (q + 1)\{-2(q + 1)^2(q^2 + 1) + p^2 \cdot 2(q - 1)^2 + 2pr(3q^2 - 2q + 6) - 8r^2q\};$$

and, thirdly, writing $a = 1$, $b = -\frac{1}{2}$, $c = -\frac{1}{2}$, we obtain

$$\Omega = (q + 1)\{(-2q^3 + q^2 - \frac{3}{2}q + \frac{3}{2}q) + p^2(-\frac{1}{2}q^2 + \frac{3}{2}q - \frac{1}{2}q) + pr(6q^2 - \frac{15}{2}q - \frac{17}{4}) + r^2(-8q + \frac{5}{2})\}.$$

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652.

ON A SEXTIC TORSE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 229–235.]

(Page 68.) THE torse having for its edge of regression or cuspidal edge the curve defined by the equations $x = \cos \phi$, $y = \sin \phi$, $z = \cos 2\phi$, is an interesting and convenient one for the construction of a model, and it is here considered partly from that point of view.

The edge is a quadriquadric curve, the intersection of the cylinder $x^2 + y^2 = 1$ with the parabolic hyperboloid $z = x^2 - y^2$; the cylinder regarded as a cone having its vertex at infinity on the line $x = 0$, $y = 0$, viz. the vertex is on the hyperboloid, or the curve is a nodal quadriquadric (the node being thus an isolated point at infinity on the line in question), and the torse is consequently of the order $8 - 2$, = 6, viz. it is a sextic torse.

The edge is a bent oval situate on the cylinder $x^2 + y^2 = 1$, such that, regarding ϕ as the azimuth (or angle measured along the circular base from its intersection with the axis of x), the altitude z is given by the equation $z = \cos 2\phi$; viz. there are in the plane xz , or, say in the planes xz , $x'z$, two maxima altitudes $z = 1$, and in the plane yz , or, say in the planes yz and $y'z$, two minima altitudes $z = -1$. The sections by these principal planes are, as is seen at once, nodal curves on the surface; they are, in fact, the cubic curves $z = 3 - \frac{2}{x^2}$, viz. here as x increases from ± 1 to $\pm \infty$, z increases from the before-mentioned value 1 to 3, and $z = -3 + \frac{2}{y^2}$, viz. as

y increases from ± 1 to $\pm \infty$, z decreases from the before-mentioned value -1 to -3 . The two half-sheets (which meet in the cuspidal edge) intersect each other along these nodal lines, in such wise that the section of the surface by any axial plane (plane through the line $x = 0, y = 0$) is a curve having a cusp on the cuspidal edge, and such that when the axial plane coincides with either of the principal planes $x = 0, y = 0$, the

(Page 69.) two half-branches of the curve coincide together with the portions which lie outside the cylinder $x^2 + y^2 = 1$, in fact, the portions referred to above, of the nodal curve in the plane in question; the portions which lie inside the cylinder are acnodal or isolated curves without any real sheet through them. It may be added, in the way of general description, that the section of the surface by any cylinder $x^2 + y^2 = c^2$ ($c > 1$) is a curve of the form $z = C \cos(2\theta \pm B)$, θ the angle along the base of the cylinder from the intersection with the axis of x ; C, B are functions of c ; viz. we have for the two half sheets respectively

$$z = C \cos(2\theta + B) \text{ and } z = C \cos(2\theta - B),$$

each curve having thus the two maxima $+C$, and the two minima $-C$; and the two curves intersect each other at the four points in the two principal planes respectively; viz. the points for which $\theta = 0, 90^\circ, 180^\circ, 270^\circ$, and $z = C \cos B, -C \cos B, C \cos B, -C \cos B$ accordingly.

Proceeding to discuss the surface analytically, we have for the equations of a generating line

$$\frac{x - \cos \phi}{-\sin \phi} = \frac{y - \sin \phi}{\cos \phi} = \frac{z - \cos 2\phi}{-2 \sin 2\phi} = \rho \text{ suppose,}$$

or say

$$\begin{aligned} x &= \cos \phi - \rho \sin \phi, \\ y &= \sin \phi + \rho \cos \phi, \\ z &= \cos 2\phi - 2\rho \sin 2\phi, \end{aligned}$$

which equations, considering therein ρ, ϕ as arbitrary parameters, determine the surface.

Writing $x = 0$, we find $y = \frac{1}{\cos \phi}$, and then $z = -3 + 2 \sin^2 \phi$, viz. we have

$$x = 0, \quad z = -3 + \frac{2}{y^2}, \text{ for section in plane } yz,$$

and, similarly, writing $y = 0$, we find $x = \frac{1}{\cos \phi}$, and then $z = 3 - 2 \cos^2 \phi$, viz.

$$y = 0, \quad z = 3 - \frac{2}{x^2}, \text{ for section by plane } xz.$$

By what precedes, these are nodal curves, crunodal for the portions

$$(y = \pm 1 \text{ to } \pm \infty, z = -1 \text{ to } -3) \text{ and } (x = \pm 1 \text{ to } \pm \infty, z = 1 \text{ to } 3)$$

respectively, acnodal for the remaining portions $y < \pm 1, x < \pm 1$ respectively.

Writing $x = r \cos \theta, y = r \sin \theta$, so that the coordinates of a point on the surface are r, θ, z , where $r = \sqrt{(x^2 + y^2)}$ is the projected distance, θ is the azimuth from the axis of x , and z is the altitude, we have

$$\begin{aligned} r \cos \theta &= \cos \phi - \rho \sin \phi, \\ r \sin \theta &= \sin \phi + \rho \cos \phi, \\ z &= \cos 2\phi - 2\rho \sin 2\phi. \end{aligned}$$

(Page 70.) We have $r^2 = 1 + \rho^2$; and thence also, if $\tan \alpha = 2\rho, = \pm 2\sqrt{(r^2 - 1)}$, that is,

$$\cos \alpha = \frac{1}{\sqrt{(4r^2 - 3)}}, \quad \sin \alpha = \pm \frac{2\sqrt{(r^2 - 1)}}{\sqrt{(4r^2 - 3)}},$$

then

$$z = \sqrt{(4r^2 - 3)} \cos(2\theta + \beta),$$

showing that for a given value of r (or section by the cylinder $x^2 + y^2 = r^2$) the maximum and minimum values of z are $z = \pm \sqrt{(4r^2 - 3)}$.

But proceeding to eliminate ϕ , we find

$$\begin{aligned} r^2 \cos 2\theta &= (1 - \rho^2) \cos 2\phi - 2\rho \sin 2\phi, \\ r^2 \sin 2\theta &= 2\rho \cos 2\phi + (1 - \rho^2) \sin 2\phi \end{aligned}$$

or multiplying these by $1 + 3\rho^2$ and 2ρ and adding

$$r^2 \{(1 + 3\rho^2) \cos 2\theta + 2\rho^2 \sin 2\theta\} = (1 + \rho^2)^2 (\cos 2\phi - 2\rho \sin 2\phi),$$

that is,

$$r^2 \{(3r^2 - 2) \cos 2\theta \pm 2(r^2 - 1)^{\frac{1}{2}} \sin 2\theta\} = r^4 z;$$

or, finally,

$$r^2 z = (3r^2 - 2) \cos 2\theta + 2(r^2 - 1)^{\frac{1}{2}} \sin 2\theta,$$

which is the equation of the surface in terms of the coordinates r, θ, z .

Observing that $(3r^2 - 2)^2 + 4(r^2 - 1)^3 = r^4(4r^2 - 3)$, we may write

$$\begin{aligned} r^2 \sqrt{(4r^2 - 3)} \cos \beta &= 3r^2 - 2, \\ r^2 \sqrt{(4r^2 - 3)} \sin \beta &= 2(r^2 - 1)^{\frac{3}{2}}, \end{aligned}$$

and therefore also

$$\tan \beta = \frac{2(r^2 - 1)^{\frac{3}{2}}}{3r^2 - 2},$$

and the equation thus becomes

$$z = \sqrt{(4r^2 - 3)} \cos(2\theta + \beta),$$

where z is the altitude belonging to the azimuth θ in the cylindrical section, radius r . The maxima and minima altitudes are $\pm \sqrt{(4r^2 - 3)}$, and these correspond to the values $\theta = \pm \frac{1}{2}\beta, \frac{1}{2}\pi + \frac{1}{2}\beta, \pi + \frac{1}{2}\beta, \frac{3}{2}\pi + \frac{1}{2}\beta$; it is to be further noticed that when $r = 1$, we have $\beta = 0$, but as r increases and becomes ultimately infinite, β increases to $\frac{1}{2}\pi$, that is, $\frac{1}{2}\beta$ increases from 0 to $\frac{1}{4}\pi$.

It may be noticed that the surface is a peculiar kind of deformation, obtained by giving proper rotations to the several cylindrical sections of the surface $z = \sqrt{(4r^2 - 3)} \cos 2\theta$; viz. in rectangular coordinates this is $r^2 z = \sqrt{(4r^2 - 3)}(x^2 - y^2)$, that is,

$$(x^2 + y^2)^4 z^2 - \{4(x^2 + y^2) - 3\}(x^2 - y^2)^2 = 0.$$

To obtain the equation in rectangular coordinates, we have

$$r^4 z^2 - 2z(3r^2 - 2)(x^2 - y^2) + (3r^2 - 2)^2 - 16(r^2 - 1)^3 \sin^2 2\theta = 0,$$

viz. this is

$$r^4 z^2 - 2z(3r^2 - 2)(x^2 - y^2) + (3r^2 - 2)^2 \left(1 - \frac{4x^2 y^2}{r^4}\right) - 16(r^2 - 1)^3 \frac{4x^2 y^2}{r^4} = 0,$$

(Page 71.) or, what is the same thing, it is

$$r^4 z^2 - 2z(3r^2 - 2)(x^2 - y^2) + (3r^2 - 2)^2 - \frac{4x^2 y^2}{r^4} \{4(r^2 - 1)^3 + (3r^2 - 2)^2\} = 0,$$

viz. the term in $\{ \}$ being $r^4(4r^2 - 3)$, this is

$$r^4 z^2 - 2z(3r^2 - 2)(x^2 - y^2) + (3r^2 - 2)^2 - 4x^2 y^2(4r^2 - 3) = 0,$$

or say

$$z^2(x^2 + y^2)^2 - 2z(3x^2 + 3y^2 - 2)(x^2 - y^2) + (3x^2 + 3y^2 - 2)^2 - 4x^2 y^2(4x^2 + 4y^2 - 3) = 0.$$

This may also be written

$$\{z(x^2 - y^2) - 3x^2 - 3y^2 + 2\}^2 + 4x^2 y^2(z^2 - 4x^2 - 4y^2 + 3) = 0,$$

a form which puts in evidence the nodal curves

$$x = 0, \quad zy^2 = -3y^2 + 2, \quad \text{and} \quad y = 0, \quad zx^2 = 3x^2 - 2.$$

It shows also that the quadric cone $z^2 - 4x^2 - 4y^2 + 3 = 0$ touches the surface along the curve of intersection with the surface $z(x^2 - y^2) - 3(x^2 + y^2) + 2 = 0$. This is, in fact, the curve of maxima and minima of the cylindrical sections, viz. reverting to the form $z = \sqrt{(4r^2 - 3)} \cos(2\theta + \beta)$, or, if for greater clearness, attending only to one sheet of the surface, we write it $z = \sqrt{(4r^2 - 3)} \cos(2\theta - \beta)$, we have a maximum, $z = \sqrt{(4r^2 - 3)}$, for $2\theta = \beta$ (or $2\pi + \beta$), giving

$$\cos 2\theta = \cos \beta = \frac{3r^2 - 2}{r^2 \sqrt{(4r^2 - 3)}} = \frac{3r^2 - 2}{r^2 z},$$

and a minimum, $z = -\sqrt{(4r^2 - 3)}$, for $2\theta = \pi + \beta$ (or $3\pi + \beta$), giving

$$\cos 2\theta = -\cos \beta = -\frac{3r^2 - 2}{r^2 \sqrt{(4r^2 - 3)}} = \frac{3r^2 - 2}{r^2 z},$$

viz. the locus is $z^2 = 4(r^2 - 3)$, $z(x^2 - y^2) = 3r^2 - 2$; and for $z = \sqrt{(4r^2 - 3)} \cos(2\theta + \beta)$ we find the same locus, viz. the equations of the locus are

$$z^2 - 4x^2 - 4y^2 + 3 = 0, \quad z(x^2 - y^2) - 3x^2 - 3y^2 + 2 = 0,$$

as above.

To put in evidence the cuspidal edge, write for a moment $\zeta = z - x^2 + y^2$, the equation becomes $\{\zeta(x^2 - y^2) + (r^2 - 1)(r^2 - 2) - 4x^2 y^2\}^2 + 4x^2 y^2\{\zeta^2 + 2\zeta(x^2 - y^2) + (r^2 - 1)(r^2 - 3) - 4x^2 y^2\} = 0$;

viz. this is

$$\zeta^2 r^4 + 2\zeta(x^2 - y^2)(r^2 - 1)(r^2 - 2) + (r^2 - 1)^2(r^2 - 2)^2 - 4\zeta^2 x^2 y^2(r^2 - 1)^2 = 0,$$

or writing the last term thereof in the form

$$-[r^4 - (x^2 - y^2)^2](r^2 - 1)^2,$$

and then putting $r^2 = 1 + U$, the equation is

$$\zeta^2(1 + 2U + U^2) + 2\zeta U(U - 1)(x^2 - y^2) + U^2(U - 1)^2 - U^2\{(U + 1)^2 - (x^2 - y^2)^2\} = 0;$$

viz. this is

$$\{\zeta - U(x^2 - y^2)\}^2 + 2U\{\zeta^2 + \zeta U(x^2 - y^2) - 2U^2\} + 4U^2 = 0,$$

(Page 72.) showing the cuspidal edge $\zeta = 0, U = 0$, viz. $z = x^2 - y^2, x^2 + y^2 = 1$. Moreover, along the cuspidal edge the surface is touched by $\zeta - U(x^2 - y^2) = 0$, that is, by $z - (x^2 - y^2) = 0$; and at the points where this tangent surface again meets the surface we have $(x^2 - y^2)^2(x^2 + y^2 + 3) - 4 = 0$; viz. the surface contains upon itself the curve represented by this last equation, and $z - (x^2 - y^2) = 0$.

As a verification, in the form

$$\{z(x^2 - y^2) - 3x^2 - 3y^2 + 2\}^2 + 4x^2y^2(z^2 - 4x^2 - 4y^2 + 3) = 0$$

of the equation of the surface, write $z = x^2 - y^2$. If for a moment $x^2 + y^2 = \lambda, x^2 - y^2 = \mu$, then the value of z is $z = \lambda\mu$, and the equation becomes

$$(\lambda\mu^2 - 3\lambda + 2)^2 + (\lambda^2 - \mu^2)(\lambda^2\mu^2 - 4\lambda + 3) = 0,$$

that is,

$$\mu^2(\lambda^4 - 6\lambda^2 + 8\lambda - 3) - 4\lambda^3 + 12\lambda^2 - 12\lambda + 4 = 0;$$

or, what is the same thing,

$$(\lambda - 1)^2\{\mu^2(\lambda + 3) - 4\} = 0,$$

so that we have $(\lambda - 1)^2 = 0$, or else $\mu^2(\lambda + 3) - 4 = 0$; viz. $(x^2 + y^2 - 1)^2 = 0$, or else $(x^2 - y^2)^2(x^2 + y^2 + 3) - 4 = 0$, agreeing with the former result.

In polar coordinates, the surface is touched along the cuspidal edge by the surface $z = r^2 \cos 2\theta$, and where this again meets the surface we have $r^4(r^2 + 3) \cos^2 2\theta - 4 = 0$.

For the model, taking the unit to be 1 inch, I suppose that for the edge of regression we have

$$x = 2 \cos \phi, \quad y = 2 \sin \phi, \quad z = 5 + (.45) \cos 2\phi,$$

viz. the curve is situate on a cylinder radius 2 inches. And I construct in zinc-plate the cylindric sections, or say the templets, for one sheet of the surface, for the several radii 2, 3, ..., 8 inches; taking the radius as k inches, the circumference of the cylinder, or entire base of the flattened templet, is $= 2k\pi$; and the altitude, writing 2θ in place of $2\theta - \beta$ as above, is given by the formula $z = 5 + (.45)\sqrt{(k^2 - 3)} \cos 2\theta$, so that the half altitude of the wave is $= (.45)\sqrt{(k^2 - 3)}$; having this value, the curve is at once constructed geometrically. We have, moreover, $\cos \beta = \frac{3k^2 - 8}{k^2\sqrt{(k^2 - 3)}}$; the numerical values then are

k	$2k\pi$	$(.45)\sqrt{(k^2 - 3)}$	$\frac{3k^2 - 8}{k^2\sqrt{(k^2 - 3)}}$	$\frac{1}{2}\beta$
2	12.57	0.45	1.00	0°
3	18.85	1.10	.86	15
4	25.13	1.62	.69	23
5	31.42	2.11	.57	27 $\frac{1}{2}$
6	37.70	2.59	.48	30 $\frac{1}{2}$
7	43.98	3.05	.42	32 $\frac{1}{2}$
8	50.27	3.51	.36	34

the altitudes in the successive templets being thus included between the limits $5 \pm 0.45, 5 \pm 1.10, \dots, 5 \pm 3.51$.

653.

ON A TORSE DEPENDING ON THE ELLIPTIC FUNCTIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 235–241.]

(Page 73.) ON attempting to cover with paper one half-sheet of the foregoing sextic torse, [652], I found that the paper assumed approximately the form of a circular annulus of an angle exceeding 360° , and this led me to consider the general theory of the construction of a torse in paper, and, in particular, to consider the torsos such that when developed into a plane the edge of regression becomes a circular arc. It is scarcely necessary to remark that, to construct in paper a circular annulus of an angle exceeding 360° , we have only to take a complete annulus, cut it along a radius, and then insert (gumming it on to the two terminal radii) a portion of an equal circular annulus; drawing from each point of the inner circular boundary a half-tangent, and considering these half-tangents as rigid lines, the paper will bend round them so as to form the half-sheet of a torse having for its edge of regression this inner boundary, which will assume the form of a closed curve with two equal and opposite maxima and two equal and opposite minima, described on a cylinder, and being approximately such as the curve given by the equations

$$x = \cos \theta, \quad y = \sin \theta, \quad z = m \cos 2\theta.$$

Considering, in general, an arc PQ (without inflexions) of any curve, and drawing at the consecutive points $P, P', P'', \&c.$ the several half-tangents $PT, P'T', P''T'', \dots$, then, considering these as rigid lines and bending the paper round them, we have the half-sheet of a torse, having for its edge of regression the curve in question now bent into a curve of double curvature. It is, moreover, clear that the edge of regression has at each point thereof the same radius of absolute curvature as the original plane curve; in fact, if in the plane curve $PP' = ds$, and the angle $T'PT$ between the consecutive half-tangents PT and $P'T'$ be $= d\phi$, these quantities ds and

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(Page 74.) $d\phi$ remain unaltered in the curve of double curvature; and the radius of absolute curvature is given by the equation $\rho d\phi = ds$. In particular when, as above, the arc is a circular one, say of radius $= a$, then, however the paper is bent, the edge of regression has at each point thereof the radius of absolute curvature $= a$.

Consider on any given surface, at a given point P thereof, and in a given direction, an element of length PP' , then (under the restrictions presently mentioned) we can determine the consecutive element $P'P''$, such that the curve $PP'P'' \dots$ shall have at P a radius of absolute curvature $= a$; in fact, r being the radius of curvature of the normal section of the surface through the element PP' , the radius of curvature of the section inclined at an angle θ to the normal section is $= r \cos \theta$; so that we have only to take the section at the inclination $\theta, = \cos^{-1} \frac{a}{r}$ to the normal section, and we have the consecutive element $P'P''$ such that the radius of absolute curvature of the curve $PP'P''$ is $= a$. The necessary restriction, of course, is that $r > a$; thus, if at the given point P the two principal radii of curvature are of the same sign (to fix the ideas, let the two principal radii and also a be each of them positive), then we may on the surface determine a direction PQ , for which the radius of curvature of the normal section is $= a$; and then the direction of the element PP' may be any direction between PQ and the direction PR , corresponding to the greatest of the two principal radii.

Having obtained the element $P'P''$, we may, if the radius of absolute curvature at P' be given, construct the next element $P'P'''$, and so on; that is to say, on a given surface starting from a given point P and given initial direction PP' , we can (under a restriction, as above, as to the curvature at the different points of the surface) construct a curve having at the successive points thereof given values of the radius of absolute curvature; viz., the value may be given either as a function of the coordinates of the point on the surface, or as a function of the length of the

curve measured say from the initial point P ; it is in this last manner that in what follows the value of the radius of absolute curvature is assumed to be given.

We may thus, taking on paper an arc PQ with its half-tangents, apply it to a given surface, the point P to a given point, and the infinitesimal arc PP' to an element PP' in a given direction from the given point; and we thus obtain the half-sheet of a torse having for its edge of regression a determinate curve upon the surface. In particular, the arc PQ may be circular of the radius a , and the surface be a circular cylinder of radius α ; and we thus obtain the torse having for edge of regression a curve on the cylinder radius α , and such that the radius of absolute curvature is at each point $= a$. There are three cases according as $\alpha > a$, $\alpha = a$, or $\alpha < a$; it is to be remarked that if $\alpha > a$, then the curve must at each point cut the generating line of the cylinder at an angle not exceeding $\cos^{-1} \frac{a}{\alpha}$, but that in the other two cases the angle may have any value whatever; and, further, that in every case when the angle is $= 0$, viz. when the curve touches a generating line of the cylinder, then the osculating plane of the curve coincides with the tangent plane of the cylinder.

(Page 75.) The analytical theory is very simple. Taking x, y, z as functions of the length s , we have

$$\left(\frac{dx}{ds}\right)^2 + \left(\frac{dy}{ds}\right)^2 + \left(\frac{dz}{ds}\right)^2 = 1;$$

the condition, which expresses that the radius of absolute curvature is $= a$, then is

$$\left(\frac{d^2x}{ds^2}\right)^2 + \left(\frac{d^2y}{ds^2}\right)^2 + \left(\frac{d^2z}{ds^2}\right)^2 = \frac{1}{a^2}.$$

By what precedes, the point (x, y, z) may be taken to be upon a given surface, say upon the cylinder $x^2 + y^2 = \alpha^2$; we may then write $x = \alpha \cos \theta$, $y = \alpha \sin \theta$. Taking instead of s any independent variable u whatever, and using accents to denote the derived functions in regard to u , the equations become

$$\begin{aligned} x'^2 + y'^2 + z'^2 &= s'^2, \\ x''^2 + y''^2 + z''^2 - s''^2 &= \frac{1}{a^2} s'^4, \\ x &= \alpha \cos \theta, \quad y = \alpha \sin \theta. \end{aligned}$$

From the last two equations we obtain

$$x'^2 + y'^2 = \alpha^2 \theta'^2, \quad x''^2 + y''^2 = \alpha^2 (\theta''^2 + \theta'^4),$$

and the first two equations thus become

$$\begin{aligned} \alpha^2 \theta'^2 + z'^2 &= s'^2, \\ \alpha^2 (\theta''^2 + \theta'^4) + z''^2 - s''^2 &= \frac{1}{a^2} s'^4, \end{aligned}$$

and from the first of these we find

$$s'' = \frac{\alpha^2 \theta' \theta'' + z' z''}{(\alpha^2 \theta'^2 + z'^2)^{\frac{1}{2}}},$$

whence the second equation is

$$\alpha^2 (\theta''^2 + \theta'^4) + z''^2 - \frac{(\alpha^2 \theta' \theta'' + z' z'')^2}{\alpha^2 \theta'^2 + z'^2} = \frac{1}{a^2} (\alpha^2 \theta'^2 + z'^2)^2,$$

or reducing, this is

$$(\alpha^2\theta'^2 + z'^2)(\theta''^2 + \theta'^4) + (z''^2\theta'^2 - 2\theta'\theta''z'z'' - \alpha^2\theta'^2\theta''^2) = \frac{1}{a^2}(\alpha^2\theta'^2 + z'^2)^2.$$

Taking θ as the independent variable, we have $\theta' = 1$, $\theta'' = 0$, and the equation becomes

$$\alpha^2 + z'^2 = \frac{1}{a^2}(\alpha^2 + z'^2)^2,$$

or, what is the same thing,

$$z''^2 = \frac{1}{a^2}(\alpha^2 + z'^2)^3 - (\alpha^2 + z'^2)^2.$$

Write here

$$\alpha^2 + z'^2 = \Omega^2,$$

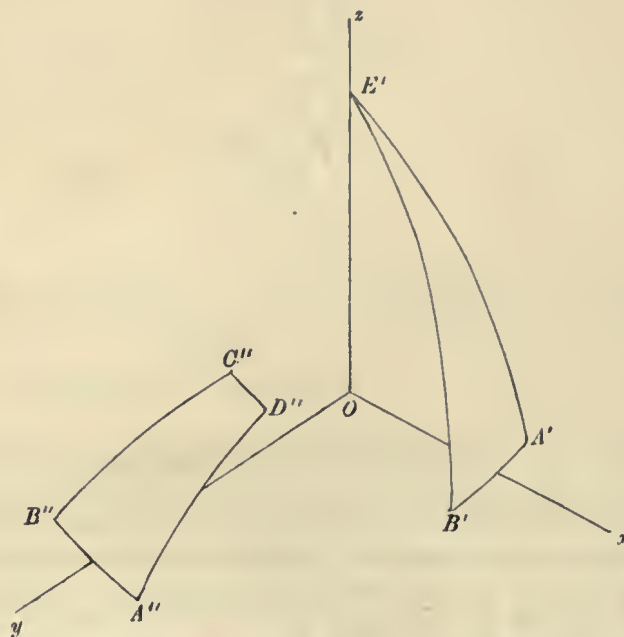
10—2

Original scan reference pages

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these is $k(1 - \cos p)$, which, however great k is, must be less than the arc of meridian $A''D'' = p$; substituting for k the value $\frac{1}{\sin p}$, the condition is $\frac{1 - \cos p}{\sin p} < p$, viz. this

Fig. 2.



is $\tan \frac{1}{2}p < p$, which is true for every value up to $p = 90^\circ$. But, more than this, we should have

$$k^2(1 - \cos p)^2 + E^2(k, p) < p^2,$$

viz. writing as before $k = \frac{1}{\sin p}$, this is

$$E^2\left(\frac{1}{\sin p}, p\right) < p^2 - \tan^2 \frac{1}{2}p;$$

this must be true, although (relating as it does to a form of E for which k is greater than 1) there might be some difficulty in verifying it.

There is, as in the first case, no limit to the value of AB , viz. this may be $= 360^\circ$, the spherical zone being then cut along a meridian, or it may be greater than 360° ; and, moreover, the spherical quadrilateral may extend south of the equator, but of course so that the limiting south latitude does not extend beyond the foregoing value $\sin^{-1} \frac{1}{k}$: viz. we may have a zone between the latitudes $\pm \sin^{-1} \frac{1}{k}$, which may be a complete zone from longitude 0° to 360° or to any greater value than 360° . The result is, that the zone is deformed into a surface of revolution, which in its general form resembles that obtained by the revolution of a half-circle or half-ellipse about a line parallel to and beyond its bounding diameter, the bounding half-diameter being less, and the greatest radius of rotation greater, than the radius of the original sphere.

642.

ON A DIFFERENTIAL RELATION BETWEEN THE SIDES OF A QUADRANGLE.

[From the *Messenger of Mathematics*, vol. VI. (1877), pp. 99—101.]

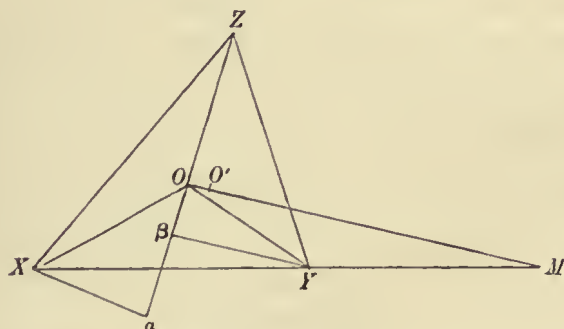
LET the sides and diagonals YZ, ZX, XY, OX, OY, OZ of a quadrangle be f, g, h, a, b, c , and let the component triangles be denoted as follows:

$$A = \triangle YZO, = (b, c, f),$$

$$B = \triangle ZXO, = (c, a, g),$$

$$C = \triangle XYO, = (a, b, h),$$

$$\Omega = \triangle XYZ, = (f, g, h),$$



viz. A, B, C, Ω are the triangles whose sides are $(b, c, f), (c, a, g), (a, b, h), (f, g, h)$ respectively, so that $\Omega = A + B + C$. Then we have between (a, b, c, f, g, h) an equation giving rise to a differential relation, which may be written

$$\Omega (Aada + Bbdb + Ccdc) - (BCfdf + CAgdg + ABhdh) = 0.$$

This may be proved geometrically and analytically. First, for the geometrical proof, it is enough to prove that, when a and b alone vary, the relation between the increments is $Aada + Bbdb = 0$; for then a and g alone varying, the relation between

the increments will be $\Omega ada - Cgdg = 0$ (as to the negative sign it is clear from the figure that a, g will increase or diminish together): and we thence at once infer the general relation.

We have consequently to prove that, considering a and b as alone variable,

$$Aada + Bbdb = 0;$$

or, what is the same thing,

$$ada : -bdb = XOZ : YOZ.$$

The points XYZ remain fixed; but O moves through the infinitesimal arc OO' , centre Z , which may be considered as situate in the right line OM drawn from O at right angles to ZO , and meeting XY produced in the point M . And then, writing for a moment $\angle OXY = X$, $\angle OYX = Y$, $\angle OMY = M$, we find at once

$$da = OO' \cos (X + M),$$

$$-db = OO' \cos (Y - M);$$

that is,

$$-\frac{da}{db} = \frac{\cos (X + M)}{\cos (Y - M)}, \text{ or } -\frac{ada}{bdb} = \frac{a \cos (X + M)}{b \cos (Y - M)}.$$

But drawing $X\alpha, Y\beta$ each of them at right angles to ZO , we have $a \cos (X + M) = X\alpha$, $b \cos (Y - M) = Y\beta$, and evidently $XOZ : YOZ = X\alpha : Y\beta$; whence the equation is $-\frac{ada}{bdb} = \frac{XOZ}{YOZ}$, which is the required relation.

For the analytical proof, it is to be observed that the relation between a, b, c, f, g, h is a quadric relation in the quantities $a^2, b^2, c^2, f^2, g^2, h^2$ respectively; this may be written

	1	a^2	a^4	
1	$b^2g^4 + b^4g^2 + c^2h^4 + c^4h^2 - (b^2 + c^2)g^2h^2 - (g^2 + h^2)b^2c^2$	$-(b^2 - c^2)(g^2 - h^2)$		
$u =$	$+(b^2 - h^2)(c^2 - g^2)$	$-b^2 - c^2 - g^2 - h^2$	$+1$	
f^2		$+1$		
f^4				$= 0 :$

say for a moment this is $A + Ba^2 + Ca^4 = 0$, where

$$\begin{aligned} A &= b^2g^4 + b^4g^2 + c^2h^4 + c^4h^2 - (b^2 + c^2)g^2h^2 - (g^2 + h^2)b^2c^2 \\ &\quad + (b^2 - h^2)(c^2 - g^2) \\ B &= -(b^2 - c^2)(g^2 - h^2) + f^2(-b^2 - c^2 - g^2 - h^2) + f^4, \\ C &= f^2; \end{aligned}$$

then we have as usual

$$\frac{du}{d \cdot a^2} ada + \frac{du}{d \cdot b^2} bdb + \&c. = 0,$$

where

$$\frac{1}{2} \frac{du}{d \cdot a^2} = Ca^2 + \frac{1}{2}B.$$

But in virtue of $u=0$, we have

$$(Ca^2 + \frac{1}{2}B)^2 = C(Ca^4 + Ba^2 + A) + \frac{1}{4}(B^2 - AC),$$

that is, $\frac{du}{d \cdot a^2} = \sqrt{(B^2 - 4AC)}$; and here $B^2 - 4AC$ is a quartic function of f^2 , which is easily seen to reduce itself to the form

$$f^2 - (g+h)^2 f^2 - (g-h)^2 f^2 - (b+c)^2 f^2 - (b-c)^2.$$

The coefficients of bdb , cdc , &c., are given as expressions of the like form; substituting their values, the differential relation is

$$\sqrt{f^2 - (g+h)^2 f^2 - (g-h)^2 f^2 - (b+c)^2 f^2 - (b-c)^2} ada + \&c. = 0,$$

which is, in fact, the foregoing result.

It is right to notice that there are in all 16 linear factors,

$$\begin{array}{cccc|l} f+g+h, & b+c+f, & c+a+g, & a+b+h & \text{say } d, f, g, h, \\ -f+g+h, & -b+c+f, & -c+a+g, & -a+b+h & d', f', g', h', \\ f-g+h, & b-c+f, & c-a+g, & a-b+h & d'', f'', g'', h'', \\ f+g-h, & b+c-f, & c+a-g, & a+b-h & d''', f''', g''', h'''; \end{array}$$

and this being so, the coefficients of ada , bdb , cdc , fdf , gdg , hdh , are

$$\begin{array}{ll} \sqrt{(dd'd''d'''.ff'f''f''')}, & -\sqrt{(gg'g''g'''.hh'h''h''')}, \\ \sqrt{(dd'd''d'''.gg'g''g''')}, & -\sqrt{(hh'h''h'''.ff'f''f''')}, \\ \sqrt{(dd'd''d'''.hh'h''h''')}, & -\sqrt{(ff'f''f'''.gg'g''g''')}, \end{array}$$

respectively.

We may imagine the quadrilateral $ZOXY$ composed of the four rods ZO , OX , XY , YZ (lengths c , a , h , f as before) jointed together at the angles, and kept in equilibrium by forces B , G acting along the diagonals $OY (=b)$, $ZX (=a)$ respectively. We have c , a , h , f given constants, and the relation $\phi(a, b, c, f, g, h) = 0$, which connects the six quantities is the relation between the two variable diagonals (g , b); by what precedes, the differential relation $\phi'g \cdot dg + \phi'b \cdot db = 0$ is equivalent to $\Omega B b db - C A g dg = 0$. By virtual velocities we have as the condition of equilibrium $Bdb + Gdg = 0$; hence, eliminating db , dg we have

$$\frac{B}{\Omega B b} = -\frac{G}{C A g};$$

or, say

$$\frac{B}{b} : -\frac{G}{g} = \frac{1}{\Delta XYO \cdot \Delta ZYO} : \frac{1}{\Delta XYZ \cdot \Delta ZXO},$$

viz. the forces, divided by the diagonals along which they act, are proportional to the reciprocals of the products of the two pairs of triangles which stand on these diagonals respectively. The negative sign shows, what is obvious, that the forces must be, one of them a pull, the other a push.

643.

ON A QUARTIC CURVE WITH TWO ODD BRANCHES.

[From the *Messenger of Mathematics*, vol. VI. (1877), pp. 107, 108.]

It is a known theorem that the branches of a plane curve are even or odd; viz. two even branches, or an even and an odd branch (whether of the same curve or of different curves) intersect in an even number (it may be 0, and this is to be understood throughout) of real points; but two odd branches (of the same curve or of different curves) intersect in an odd number of real points*.

In particular, a right line is an odd branch, and hence it meets any even branch of a curve in an even number of real points, and an odd branch in an odd number of real points; or (what is the same thing) an even branch is one which is met by any right line whatever in an even number of real points; and an odd branch is one that is met by any right line whatever in an odd number of real points.

It is to be observed, that the simple term branch is used to denote what has been called a *complete* branch, viz. the partial branches which touch an asymptote at its opposite extremities are considered as parts of one and the same branch, and so in other cases. Thus a quadric curve, whether ellipse, parabola, or hyperbola, is one even branch; a cubic curve is either one odd branch, or else it is an odd branch and an even branch; and generally a curve of an odd order has always an odd number of odd branches, and a curve of an even order has always an even number of odd branches.

A curve without nodes has at most one odd branch; for if there were two, these would intersect in a real point, which would be a real node on the curve. In particular, a quartic curve having two odd branches must have a real node; this however may be, as in the instance about to be given, a node at infinity.

A simple instance of a quartic curve with two odd branches is that represented by the equation

$$(x^2 - 1)(y^2 + 1) - 2mxy = 0,$$

* The two branches must be distinct branches; a branch whether odd or even does not of necessity intersect itself (have upon it any real node), but it may intersect itself in an odd, or an even, number of real points.

or, what is the same thing,

$$y = \frac{1}{x^2 - 1} \left[mx \pm \sqrt{\left\{ - (x^2 - \alpha^2) \left(x^2 - \frac{1}{\alpha^2} \right) \right\}} \right],$$

where

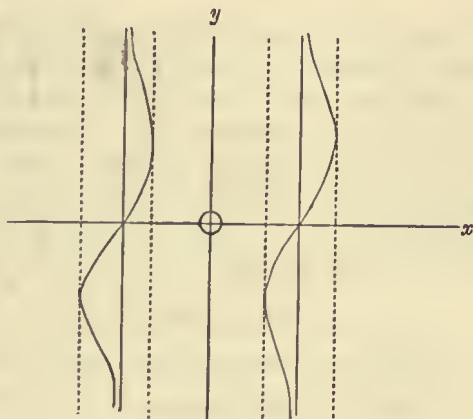
$$\alpha^2 + \frac{1}{\alpha^2} = 2 + m^2,$$

or say

$$\alpha^2 = \frac{1}{2} \{ 2 + m^2 + m \sqrt{4 + m^2} \},$$

$$\frac{1}{\alpha^2} = \frac{1}{2} \{ 2 + m^2 - m \sqrt{4 + m^2} \},$$

so that m being positive $\alpha > 1$, and the curve consists of two real branches included between the lines $x = \alpha$, $x = \frac{1}{\alpha}$, and the lines $x = -\alpha$, $x = -\frac{1}{\alpha}$ respectively; each of these lines touches the curve in a real point, viz. x having any one of the last-mentioned values, the value of y at the point of contact is $y = \frac{mx}{x^2 - 1}$; and between each pair of lines we have the asymptote $x = +1$ or $x = -1$. Hence the curve has the form shown in the figure, and it is thereby evident, that each branch of the curve is met by



any real right line whatever in one real point, or else in three real points. The numerical values in the figure are $\alpha = \frac{3}{2}$, $m = \frac{5}{6}$, whence also $x = \alpha$ or $-\frac{1}{\alpha}$, $y = 1$, and $x = -\alpha$ or $\frac{1}{\alpha}$, $y = -1$.

The curve has two nodes at infinity, viz. writing the equation in the form

$$(x^2 - z^2)(y^2 + z^2) - mxyz^2 = 0,$$

that is,

$$x^2y^2 + z^2(x^2 - y^2 - mxy) + z^4 = 0,$$

it appears that the points $(z = 0, x = 0)$, $(z = 0, y = 0)$ are each of them a node. The first of these $(z = 0, x = 0)$ is the real intersection of the two odd branches: the other of them is a conjugate point.

644.

NOTE ON MAGIC SQUARES.

[From the *Messenger of Mathematics*, vol. VI. (1877), p. 168.]

IN a magic square of any odd order, formed according to the ordinary process, there is a tolerably simple analytical expression for the number which occupies any given compartment; thus taking the square of 21, let the dexter diagonals (N.W. to S.E.) commencing from the N.E. corner compartment, be numbered 1, 2, 3, ..., 20, 21, 20', 19', ..., 2', 1', the diagonals of course containing these numbers of compartments respectively; and in any diagonal let the compartments reckoning from the top line be numbered 1, 2, 3, ..., respectively; then if $D_{\theta, \phi}$ (or $D'_{\theta, \phi}$ as the case may be) denotes the number in the compartment ϕ of the diagonal θ or θ' , we have

$$D_{2\theta+1, \phi} = 20\theta + 10 + \phi,$$

$$D_{2\theta, \phi} = 20\theta + 231 + \phi(-21),$$

$$D'_{2\theta+1, \phi} = -22\theta + 430 + \phi,$$

$$D'_{2\theta, \phi} = -22\theta + 231 + \phi(-21),$$

where in the second and fourth expressions the term -21 is to be retained only if $\phi > \theta$; if $\phi \not> \theta$, it is to be omitted. There would be a like formulæ for a square of any odd order, and it would be easy to write down the formulæ for the general value $2n+1$: but I have preferred to give them for a specific case.

645.

A SMITH'S PRIZE PAPER, 1877.

[From the *Messenger of Mathematics*, vol. VI. (1877), pp. 173—182.]

THE paper was as follows:

1. Show (independently of the theory of roots) how, if x satisfies an equation of the order n , a given rational function of x can in general be expressed as a rational and integral function of the order $n-1$. State the theorem in a more precise form, so as to make it true universally.

2. Investigate the form of the factors of $1 \pm \sin(2n+1)x$ considered as a function of $\sin x$; and give the formulæ in the two cases, $2n+1=3$ and 5 respectively.

3. Write down the substitutions which do not alter the function $ab+cd$; and explain the constitution of the group.

4. Find in a form adapted for calculation an approximate value for the sum of the middle $2\alpha+1$ terms of the expansion of $(1+1)^n$, n being a large number, and α small in comparison therewith.

Obtain thence a complete and precise statement of the theorem that in a large number of tosses the numbers of heads and tails will probably be nearly equal.

5. A point in space is represented on a given plane by its projections from two fixed points. Show how a problem relating to points, lines, and planes, is thereby reduced to a problem *in plano*; and apply the method to construct the line of intersection of two planes each passing through three given points.

6. A weight is supported on a tripod of three unequal legs resting on a smooth horizontal plane, their feet connected in pairs by strings of given lengths. Show how to determine the tensions of the several strings.

7. Explain the ordinary configuration of a system of isoparametric lines on a spherical surface; for instance, what is the configuration when there are two points of minimum value, and one point of maximum value, of the parameter?

8. Find the attraction of an infinite circular cylinder, of uniform density, on a given exterior or interior point.

9. Determine the number of arbitrary constants contained in the equation of a surface of the order r which passes through the curve of intersection of two given surfaces of the orders m and n respectively.

10. Find, for the several values of p , the number of the conics passing through p given points and touching $5-p$ given lines; and, in each case, show how to obtain (in point-coordinates or line-coordinates, as may be most simple) the equations of the conics satisfying the conditions in question.

11. Investigate the theory of the linear transformation of a ternary quadric function into itself.

12. Explain the theory of the solution of a partial differential equation, given function of x, y, z, p, q, r = arbitrary constant H ; where p, q, r are the differential coefficients of the dependent variable u in regard to the independent variables x, y, z respectively.

I propose, not (as in former years) to give complete solutions, but only to notice in more or less detail the leading points in the several questions.

1. The expression is of course required, not only for a given integral function of x , but for a given fractional function. The case where the given function is integral presents no difficulty; when the given function is fractional, the most simple case is where it is $= \frac{1}{x-a}$; supposing the equation to be $f(x)=0$, here dividing $f(x)$ by $x-a$, we have a quotient $R(x)$ which is a rational and integral function of an order not exceeding $n-1$, and a remainder which is $=f(a)$; that is,

$$\frac{f(x)}{x-a} = R(x) + \frac{f(a)}{x-a};$$

or, in virtue of the given equation $\frac{f(a)}{x-a} = -R(x)$, viz. we have thus $\frac{1}{x-a}$ in the required form. But if $f(a)=0$, then we do not obtain such an expression of $\frac{1}{x-a}$.

It has to be shown that the like considerations apply to any fractional function, and the precise form of the theorem is, that any rational function of x which does not become infinite for any value of x satisfying the given equation, can be expressed as a rational and integral function of an order not exceeding $n-1$.

2. The function $1 - \sin(2n+1)x$ is a rational and integral function of $\sin x$, of the order $2n+1$; which if n is even (or $2n+1=4p+1$) contains, as is at once seen, the factor $1 - \sin x$, but if n is odd (or $2n+1=4p-1$) the factor $(1 + \sin x)$. Suppose that any other factor is $1 - \frac{\sin x}{\sin \alpha}$, where $\sin \alpha$ not $= \pm 1$; then this will be a double factor if only $\sin x = \sin \alpha$ satisfies the condition

$$0 = \frac{d}{d \sin x} \{1 - \sin(2n+1)x\},$$

that is, $0 = \frac{\cos(2n+1)x}{\cos x}$; the value in question gives $\sin(2n+1)x = 1$, and therefore $\cos(2n+1)x = 0$; and it does not give $\cos x = 0$; hence every such factor $1 - \frac{\sin x}{\sin \alpha}$ is a double factor, or we have

$$1 - \sin(2n+1)x = (1 \pm \sin x) \Pi \left(1 - \frac{\sin x}{\sin \alpha}\right)^2.$$

Or the like result might be obtained by considering instead of $1 - \sin(2n+1)x$, the more general function

$$\sin(2n+1)a \pm \sin(2n+1)x,$$

and finally assuming $a = \frac{1}{2}\pi$.

3. Relates to a theory which is not, but ought to be, treated of in the text books of the University. See Serret's *Algèbre Supérieure*, t. II., Sect. IV.

The substitutions which leave $ab + cd$ unaltered are

1	1, that is, the letters remain unchanged,
α	(ab) , that is, a and b are interchanged,
β	(cd) , that is, c and d are interchanged,
γ	$(ab)(cd)$, that is, a and b and also c and d are simultaneously interchanged,
δ	$(ac)(bd)$, same with a and c , b and d ,
ϵ	$(ad)(bc)$, same with a and d , b and c ,
ζ	$(acbd)$, that is, we cyclically change a into c , c into b , b into d , and d into a ,
θ	$(adbc)$, that is, we cyclically change a into d , d into b , b into c , and c into a ,

viz. we have eight substitutions 1, α , β , γ , δ , ϵ , ζ , θ forming a group; that is, the product of any two of them, in either order, is a substitution of the group (or, what is the same thing, the effect of the successive performance of the two upon any arrangement $abcd$ is the same as that of the performance thereon of some other substitution of the group); thus we have $\alpha^2 = 1$, $\beta^2 = 1$, $\gamma^2 = 1$, $\alpha\beta = \beta\alpha = \gamma$, &c.; the system of these equations, which verify that the set of substitutions form a group, defines the constitution of the group—thus to take a more simple instance, a group of 4 may be 1, α , α^2 , α^3 ($\alpha^4 = 1$) or 1, α , β , $\alpha\beta$, ($\alpha^2 = 1$, $\beta^2 = 1$, $\alpha\beta = \beta\alpha$).

4. The expression of the general coefficient is

$$= \frac{1 \cdot 2 \dots 2n}{1 \cdot 2 \dots n - \alpha \cdot 1 \cdot 2 \dots n + \alpha},$$

which can be transformed by the well-known formula

$$1 \cdot 2 \dots n = n^{n+\frac{1}{2}} \sqrt{(\pi)} e^{-n},$$

viz. the coefficient thus becomes

$$= \frac{2^{2n}}{\sqrt{(n\pi)}} \frac{1}{\left(1 - \frac{\alpha}{n}\right)^{n-\alpha+\frac{1}{2}} \left(1 + \frac{\alpha}{n}\right)^{n+\alpha+\frac{1}{2}}}.$$

Now α is supposed small in comparison with n , and the factors in the denominator have the logarithms

$$(n - \alpha + \tfrac{1}{2}) \log \left(1 - \frac{\alpha}{n}\right), = (n - \alpha + \tfrac{1}{2}) \left(-\frac{\alpha}{n} + \tfrac{1}{2} \frac{\alpha^2}{n^2}\right), = -\alpha + \tfrac{1}{2} \frac{\alpha^2}{n},$$

and

$$(n + \alpha + \tfrac{1}{2}) \log \left(1 + \frac{\alpha}{n}\right), = \alpha + \tfrac{1}{2} \frac{\alpha^2}{n};$$

hence the denominator is $= e^{\frac{\alpha^2}{n}}$, and the final approximate value of the coefficient is

$$= \frac{2^{2n}}{\sqrt{(n\pi)}} e^{-\frac{\alpha^2}{n}}.$$

Hence, converting as usual the sum into a definite integral, we have the sum of the $2\alpha + 1$ coefficients

$$= \frac{2^{2n}}{\sqrt{(n\pi)}} \int_{-1}^1 e^{-\frac{\alpha^2}{n}} d\alpha,$$

or, what is the same thing,

$$= \frac{2^{2n}}{\sqrt{(\pi)}} \int_{-\frac{\alpha}{\sqrt{n}}}^{\frac{\alpha}{\sqrt{n}}} e^{-x^2} dx.$$

For the chance that the number of tosses lies between $n + \alpha$ and $n - \alpha$, this has merely to be divided by 2^{2n} ; hence writing $\alpha = kn$, the chance that the number may be between $n(1 + k)$ and $n(1 - k)$ is

$$= \frac{1}{\sqrt{(\pi)}} \int_{-k\sqrt{n}}^{k\sqrt{n}} e^{-x^2} dx,$$

where observe that the integral, taken with the limits $\infty, -\infty$ has the value $\sqrt{(\pi)}$.

Considering k as a given fraction however small, by increasing n we make $k\sqrt{n}$ as large as we please, and therefore the integral, as nearly as we please $= \sqrt{(\pi)}$, or the chance as nearly as we please $= 1$; and hence the complete and precise statement of the theorem, viz. by sufficiently increasing the number of tosses, the probability that the deviation from equality shall be any given percentage (as small as we please) of the whole number of tosses, can be made as nearly as we please equal to certainty.

Further, restoring α in place of kn , the chance of a number between $n + \alpha$ and $n - \alpha$ is

$$= \frac{1}{\sqrt{(\pi)}} \int_{-\frac{\alpha}{\sqrt{n}}}^{\frac{\alpha}{\sqrt{n}}} e^{-x^2} dx,$$

which when $\frac{\alpha}{\sqrt{(n)}}$ is small is $= \frac{2\alpha}{\sqrt{(n\pi)}}$, (more accurately $\frac{2\alpha+1}{\sqrt{(n\pi)}}$, when α is small); hence, however large α is, the chance of a deviation from equality not exceeding $\pm \alpha$, continually diminishes with n , and by making n sufficiently large becomes as small as we please.

5. The point is represented in the given plane by two points which lie *in line* with a fixed point (say O) of that plane, viz. O is the intersection of the given plane by the line which joins the two projecting points.

A line is represented on the given plane by two lines, viz. these are the projections of the line from the two given points; each point of the line is represented by the points of intersection of the two lines by any line through O .

A plane may be represented on the given plane by means of its trace thereon, and of the two points (*in line* with O) which represent any point of the plane.

Thus any problem relating to points, lines, and planes, in space is converted into a problem of plane geometry. For instance, to find the trace on the given plane of a plane through three given points A, B, C , the three points are represented by means of the pairs of points $A_1, A_2; B_1, B_2; C_1, C_2$, the points of each pair lying *in line* with O ; the required trace passes through the intersections with the given plane of the lines BC, CA, AB respectively, and we hence find it as the line through the three points which are the intersections of B_1C_1, B_2C_2 , of C_1A_1, C_2A_2 , and of A_1B_1, A_2B_2 respectively; that these points are in a line is a theorem of plane geometry, which, if not previously known, would have at once been given by the construction.

6. The solution ought obviously to be obtained from the principle of virtual velocities; taking a, b, c for the lengths of the legs, f, g, h for the lengths of the strings, and z for the height of the summit, z is a known function of a, b, c, f, g, h , (z is in fact $= \frac{3V}{\Delta}$, where V , the volume of the tetrahedron, is a given function of a, b, c, f, g, h ; and Δ , the area of the base, is a given function of f, g, h). Writing then F, G, H for the tensions, and W for the weight, and regarding z, f, g, h as variable, the principle gives

$$Wdz + Fdf + Gdg + Hdh = 0,$$

that is,

$$F, G, H, = -W \frac{dz}{df}, -W \frac{dz}{dg}, -W \frac{dz}{dh},$$

respectively.

7. The ordinary case is when an isoparametric line has on one side of it larger values, on the other side of it smaller values of the parameter; the case where the isoparametric line is a line of maximum, or of minimum, parameter is excluded.

The lines in the neighbourhood of a point of maximum, or of minimum, parameter are ovals surrounding the point in question, each oval being itself surrounded by the consecutive oval. Supposing that there are two points of minimum parameter, we have round each of them a series of ovals, until at length an oval belonging to the one of them comes to unite itself with an oval belonging to the other, the two ovals altering themselves into a figure of eight. Surrounding this we have a closed curve (in the first instance a deeply twice-indented oval) which (in the case supposed of there being, besides the two points of minimum parameter, a single point of maximum parameter) is in fact an oval surrounding the point of maximum parameter, and the remaining curves are the series of ovals surrounding that point. If we project stereographically from the point of maximum parameter (so that this point is represented by the points at infinity) we have a figure of eight, each loop containing within it a series of continually diminishing closed curves, and the figure of eight itself surrounded by a series of continually increasing closed curves.

8. The investigation by means of the Potential presents the difficulty that the Potential of the infinite cylinder has no determinate value, as at once appears from the limiting case where the cylinder is reduced to a right line; the difficulty is perhaps rather apparent than real, inasmuch as the partial differential equations contain only differential coefficients $\frac{dV}{dr}$, $\frac{d^2V}{dr^2}$, where $\frac{dV}{dr}$ as representing an attraction, and therefore also $\frac{d^2V}{dr^2}$, are determinate. But it is safer to work directly with the Attraction; the Attraction of an infinite line acts in the perpendicular plane through the attracted point, and is inversely proportional to the distance; the problem is thus reduced to the plane problem of a circle of uniform density, force varying as (distance)⁻¹, attracting a point in its own plane. This is precisely similar to the case of a sphere with the ordinary law of attraction; dividing the circle into rings, each ring exerts an attraction = 0 upon an interior point, and an attraction as if collected at the centre upon an exterior point. Hence, writing a for the radius of the cylinder, and r for the distance of the attracted point, the attraction is $=\pi r$ for an interior point, and $=\frac{\pi a^2}{r}$ for an exterior point.

9. The theory is precisely the same as for curves; taking the surfaces to be $U=0$ of the order m , and $V=0$ of the order n , the general form of the equation of a surface of the order r (r not less than m or n) is $LU+MV=0$, where L is the general function of the order $r-m$, and M the general function of the order $r-n$; and so long as r is less than $m+n$, we obtain the required number of arbitrary constants as the sum of the numbers of the coefficients of L and of M , less unity. But as soon as r is $=m+n$ a modification arises, viz. we obtain here an identity by assuming $L=V$, $M=-U$, and so for any larger value of r , we have an identity by assuming $L=V\phi$, $M=-U\phi$, where ϕ is the general function of the order $r-m-n$.

10. The numbers are known to be 1, 2, 4, 4, 2, 1, which values are obtained most easily (though not in the way which is theoretically most interesting) by finding for

the first three cases the equation of the required conic in point-coordinates; and then, by changing these into line-coordinates, we have the equations for the remaining three cases.

$p = 5$: 5 points. The equation of the conic is

$$(a, b, c, f, g, h \chi x, y, z)^2 = 0,$$

and we have 5 linear equations to determine the ratios of the coefficients; the number is therefore = 1.

$p = 4$: 4 points and 1 line. Taking $U = 0$ and $V = 0$, the equations of any two conics each passing through the four points, the equation of the required conic will be $U + \lambda V = 0$, and the condition of touching a given line gives a quadric equation for λ ; the number is therefore = 2.

$p = 3$: 3 points and 2 lines. In the same manner, by taking $U = 0$, $V = 0$, $W = 0$, for the equations of any three conics through the three points; or if the equations of the lines containing the three points in pairs are $x = 0$, $y = 0$, $z = 0$, then the equations of the three conics are $yz = 0$, $zx = 0$, $xy = 0$, and the equation of any conic through these points is $fyz + gzx + hxy = 0$; the conditions of touching two given lines $\xi x + \eta y + \zeta z = 0$ and $\xi' x + \eta' y + \zeta' z = 0$, are

$$\sqrt{f} \sqrt{\xi} + \sqrt{g} \sqrt{\eta} + \sqrt{h} \sqrt{\zeta} = 0, \quad \sqrt{f} \sqrt{\xi'} + \sqrt{g} \sqrt{\eta'} + \sqrt{h} \sqrt{\zeta'} = 0;$$

we have thus the ratios $\sqrt{f} : \sqrt{g} : \sqrt{h}$ linearly determined in terms of $\sqrt{\xi}$, $\sqrt{\eta}$, &c.; there is no loss of generality in taking $\sqrt{\xi}$, $\sqrt{\xi'}$ each with a determinate sign, the signs of $\sqrt{\eta}$, &c. being then arbitrary, we have $2^4 = 16$ values of $\sqrt{f} : \sqrt{g} : \sqrt{h}$, and therefore one-fourth of this = 4, for the number of values of $f : g : h$; that is, the number is = 4.

11. This is a known theory; taking x_1, y_1, z_1 for the linear functions of x, y, z , which are such that

$$(a, b, c, f, g, h \chi x_1, y_1, z_1)^2 = (a, b, c, f, g, h \chi x, y, z)^2,$$

then assuming $x_1, y_1, z_1 = 2\xi - x, 2\eta - y, 2\zeta - z$ respectively, we have

$$(a, \dots \chi 2\xi - x, 2\eta - y, 2\zeta - z)^2 = (a, \dots \chi x, y, z)^2,$$

or, omitting terms which destroy each other, and throwing out the factor 4, this is

$$(a, \dots \chi \xi, \eta, \zeta)^2 = (a, \dots \chi \xi, \eta, \zeta \chi x, y, z),$$

an equation which is satisfied identically by assuming

$$\begin{aligned} a\xi + h\eta + g\zeta &= ax + hy + gz & . & - \nu\eta + \mu\zeta, \\ h\xi + b\eta + f\zeta &= hx + by + fz + \nu\xi & . & - \lambda\zeta, \\ g\xi + f\eta + c\zeta &= gx + fy + cz - \mu\xi + \lambda\eta & . & . \end{aligned}$$

where λ, μ, ν are arbitrary; viz. multiplying by ξ, η, ζ , and adding we have the equation in question. The three equations determine ξ, η, ζ as linear functions of x, y, z ; and we have thence x_1, y_1, z_1 as linear functions of x, y, z ; viz. this is a solution containing three arbitrary constants λ, μ, ν .

12. The partial differential equation might equally well have been proposed in the form, given function of $x, y, z, p, q, r=0$, viz. the equation then is $\phi(x, y, z, p, q, r)=0$, the general partial differential equation involving the three independent variables x, y, z , and the derived functions p, q, r of the dependent variable u , *but not involving the dependent variable u* . The question is therefore in effect as follows: to find p, q, r functions of x, y, z connected by the foregoing equation, and, moreover, such that $pdx + qdy + rdz$ is an exact differential; for then writing $u = \int(pdx + qdy + rdz)$, we have the solution of the given partial differential equation.

Whatever be the method adopted, it comes out that the solution depends on the integration of the system of ordinary differential equations

$$\frac{dp}{\frac{d\phi}{dx}} = \frac{dq}{\frac{d\phi}{dy}} = \frac{dr}{\frac{d\phi}{dz}} = \frac{dx}{\frac{d\phi}{dp}} = \frac{dy}{\frac{d\phi}{dq}} = \frac{dz}{\frac{d\phi}{dr}},$$

and the answer consists first in showing this, and secondly, in showing how from an integral or integrals of the system we pass to the solution of the partial differential equation.

Considering the partial differential equation in the form actually proposed, we may instead of ϕ write H , where H will stand for that given function of x, y, z, p, q, r which is the value of the arbitrary constant H ; making this change, and putting the foregoing equal quantities equal to the differential dt of a new variable, the system of ordinary differential equations is

$$\begin{aligned} \frac{dp}{dt} &= -\frac{dH}{dx}, & \frac{dq}{dt} &= -\frac{dH}{dy}, & \frac{dr}{dt} &= -\frac{dH}{dz}, \\ \frac{dx}{dt} &= \frac{dH}{dp}, & \frac{dy}{dt} &= \frac{dH}{dq}, & \frac{dz}{dt} &= \frac{dH}{dr}, \end{aligned}$$

where H is a given function of x, y, z, p, q, r . This is, in fact, the Hamiltonian system of equations; and it was in view to the connexion that the partial differential equation was proposed in its actual form.

646.

ON THE GENERAL EQUATION OF DIFFERENCES OF THE SECOND ORDER.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XIV. (1877), pp. 23—25.]

CONSIDER the equation of differences

viz. we have

$$u_x = a_{x-1} u_{x-1} + b_{x-2} u_{x-2},$$

$$u_2 = a_1 u_1 + b_0 u_0,$$

$$u_3 = a_2 u_2 + b_1 u_1,$$

$$u_4 = a_3 u_3 + b_2 u_2,$$

$$u_5 = a_4 u_4 + b_3 u_3,$$

$$u_6 = a_5 u_5 + b_4 u_4,$$

&c.,

and thence

$$u_3 = \left| \begin{array}{c} a_2 a_1 \\ + b_1 \end{array} \right| u_1 + \left| \begin{array}{c} a_2 \\ b_0 \end{array} \right| u_0,$$

$$u_4 = \left| \begin{array}{c} a_3 a_2 a_1 \\ + u_3 b_1 \\ + a_1 b_2 \end{array} \right| u_1 + \left| \begin{array}{c} a_3 a_2 \\ + b_2 \end{array} \right| u_0,$$

$$u_5 = \left| \begin{array}{c} a_4 a_3 a_2 a_1 \\ + a_4 a_3 b_1 \\ + a_4 a_1 b_2 \\ + a_2 a_1 b_3 \\ + b_1 b_3 \end{array} \right| u_1 + \left| \begin{array}{c} a_4 a_3 a_2 \\ + a_4 b_2 \\ + a_2 b_3 \end{array} \right| u_0,$$

$$\begin{array}{rcl}
 u_6 = & a_5 a_4 a_3 a_2 a_1 & | \quad u_1 + \quad a_5 a_4 a_3 a_2 \quad | \quad b_0 u_0, \\
 & + a_5 a_4 a_3 b_1 & \quad \quad + a_5 a_4 b_2 \\
 & + a_5 a_4 a_1 b_2 & \quad \quad + a_5 a_2 b_3 \\
 & + a_5 a_2 a_1 b_3 & \quad \quad + a_3 a_2 b_4 \\
 & + a_3 a_2 a_1 b_4 & \quad \quad + b_4 b_2 \\
 & + a_5 b_3 b_1 & \\
 & + a_3 b_4 b_1 & \\
 & + a_1 b_4 b_2 & \\
 & & \&c.
 \end{array}$$

It is now easy to see the law; viz. writing for instance

$$u_6 = 54321 \cdot u_1 + 5432 \cdot b_0 u_0,$$

then 54321 has a leading term $a_5 a_4 a_3 a_2 a_1$: it has terms derived from this by changing any pair $a_2 a_1$ into b_1 , $a_3 a_2$ into b_2 , $a_4 a_3$ into b_3 , $a_5 a_4$ into b_4 : it has terms derived by changing any two pairs $a_4 a_3$, $a_2 a_1$ into $b_3 b_1$; $a_5 a_4$, $a_2 a_1$ into $b_4 b_1$; $a_5 a_4$, $a_3 a_2$ into $b_4 b_2$, and so on; where observe that the expression a pair denotes the product of two consecutive a 's.

And, similarly, 5432 has a leading term $a_5 a_4 a_3 a_2$; the other terms being derived from this in the same manner precisely.

The solution of $u_x = l x (a u_{x-1} - u_{x-2})$ is included in, and might be deduced from the foregoing, but it is convenient to obtain it separately. Supposing for greater simplicity that $u_{-1} = 0$, $u_0 = 1$ (or, what is the same thing, $u_0 = 1$, $u_1 = l_1 a$), then we find

$$\begin{array}{l}
 u_0 = 1, \\
 u_1 = l_1 a, \\
 u_2 = l_2 l_1 a^2 - l_2, \\
 u_3 = l_3 l_2 l_1 a^3 - \left| \begin{array}{c} l_3 l_2 \\ + l_3 l_1 \end{array} \right| a, \\
 u_4 = l_4 l_3 l_2 l_1 a^4 - \left| \begin{array}{c} l_4 l_3 l_2 \\ + l_4 l_3 l_1 \\ + l_4 l_2 l_1 \end{array} \right| a^2 + l_4 l_2, \\
 u_5 = l_5 l_4 l_3 l_2 l_1 a^5 - \left| \begin{array}{c} l_5 l_4 l_3 l_2 \\ + l_5 l_4 l_3 l_1 \\ + l_5 l_4 l_2 l_1 \\ + l_5 l_3 l_2 l_1 \end{array} \right| a^3 + \left| \begin{array}{c} l_5 l_4 l_2 \\ + l_5 l_3 l_2 \\ + l_5 l_3 l_1 \end{array} \right| a, \\
 \&c.,
 \end{array}$$

viz. we may for example write

$$u_3 = l_5 4321 \cdot a^5 - 4321 (\cdot) a^3 + 4321 (:) a;$$

where

$$4321 \text{ denotes } l_4 l_3 l_2 l_1;$$

in 4321 ($\cdot$), we omit successively each number, viz. we thus obtain

$$\begin{aligned} & 432 + 431 + 421 + 321, \\ & = l_4 l_3 l_2 + l_4 l_3 l_1 + l_4 l_2 l_1 + l_3 l_2 l_1; \end{aligned}$$

in 4321 ($:$), we omit successively each two non-consecutive numbers, viz. the omitted numbers being 1, 3; 1, 4; 2, 4, we obtain

$$\begin{aligned} & 42 + 32 + 31, \\ & = l_4 l_2 + l_3 l_2 + l_3 l_1; \end{aligned}$$

and so on, the omissions being each three numbers, each four numbers, &c., no two of them being consecutive; thus in 654321 ($\cdot$), the omissions are 5, 3, 1, and 6, 4, 2; or the symbol is

$$\begin{aligned} & 642 + 531, \\ & = l_6 l_4 l_2 + l_5 l_3 l_1. \end{aligned}$$

As an application, a solution of the differential equation $\frac{d}{dx} \left(x \frac{dy}{dx} \right) + (x - a) y = 0$ is $y = u_0 + u_1 x + u_2 x^2 + \&c.$, where $n^2 u_n = a u_{n-1} - u_{n-2}$, and in particular $1^2 u_1 = a u_0$; the equation of differences is thus of the form in question, and retaining l_n in place of its value, $= n^2$, the solution is $u_0 = 1$, $u_1 = l_1 a$, $u_2 = l_2 l_1 a^2 - l_2$, &c. *ut supra*. The differential equation was considered by the Rev. H. J. Sharpe, who mentioned it to Prof. Stokes.

647.

ON THE QUARTIC SURFACES REPRESENTED BY THE EQUATION,
SYMMETRICAL DETERMINANT = 0.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XIV. (1877),
pp. 46—52.]

CONSIDER the equation

$$\nabla = \begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & d \end{vmatrix} = 0,$$

where for the moment $(a, b, \dots)$ denote linear functions of the coordinates (x, y, z, w) . This is a quartic surface having 10 nodes; viz. if we write $(A, B, \dots)$ for the first minors of the determinant, then the cubic surfaces $A = 0, B = 0, \dots$ have in common 10 points which are nodes of the quartic surface.

Suppose that (a, b, c, f, g, h) are linear functions of the form (x, y) ; then, observing that every term of ∇ contains as a factor

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix},$$

or one of its first minors, it is clear that the line $x = 0, y = 0$ is a double line on the surface. But the number of nodes is now less than 10; in fact, writing $(x = 0, y = 0)$, we make each of the first minors of ∇ to vanish; that is, the cubic surfaces, which by their intersection determine the nodes, have in common the line $(x = 0, y = 0)$, and there is a diminution in the number of their common intersections. I do not pursue the enquiry, but pass to a different question.

I, in fact, take the terms ($a, \dots$) of the determinant to be homogeneous functions of (x, y, z, w) of the degrees

$$\begin{vmatrix} 0, & 1, & 1, & 0 \\ 1, & 2, & 2, & 1 \\ 1, & 2, & 2, & 1 \\ 0, & 1, & 1, & 0 \end{vmatrix},$$

respectively, viz. a, d, l are constants, g, h, m, n linear functions, and b, c, f quadric functions of the coordinates; $\nabla = 0$ still represents a quartic surface; and it appears by a general formula that the number of nodes is $= 8$. But we can easily show this directly; and further, that the 8 nodes are the intersections of three quadric surfaces; or say that the quartic surface is *octadic*. For denoting as before the first minors by $A, \dots$, then B, C, F are each of them a quadric function of the coordinates, viz. we have

$$B = d(ac - g^2) - cl^2 - an^2 + 2gln,$$

$$C = d(ab - h^2) - am^2 - bl^2 + 2hlm,$$

$$F = d(gh - af) + l^2f + mna - nlh - lmg,$$

and we have identically

$$BC - F^2 = (ad - l^2) \nabla,$$

so that throwing out the constant factor $ad - l^2$, the equation of the surface is

$$BC - F^2 = 0,$$

and it has 8 nodes, the intersections of the three quadric surfaces $B = 0, C = 0, F = 0$. By equating to zero any other minor of the determinant ∇ , we have a surface passing through the 8 nodes; we have for instance the quartic surface

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix} = 0.$$

Suppose now (and in all that follows) that, the degrees being as already mentioned, we further assume that b, c, f are quadric functions of the form $(x, y)^2$, g, h linear functions of the form (x, y) ; then since each term of ∇ contains either

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix}$$

or one of its first minors, it is clear that the line $(x = 0, y = 0)$ is a double line on the surface. But in the present case there is not any diminution in the number of the nodes; in fact, writing $x = 0, y = 0$, and therefore b, c, f, g, h each $= 0$ (but not $a = 0$), the minors B, C, F none of them vanish; that is, the line $x = 0, y = 0$ is not a line on any one of the quadric surfaces, and the quadric surfaces intersect as before in an octad of points.

The equation $\nabla = 0$ thus represents a quartic surface having a double line, and also 8 nodes forming an octad.

We may without loss of generality write $d = 0$; in fact, the determinant is unaltered if we add to the fourth column θ times the first column, and then to the fourth line θ times the first line; the determinant is thus of the original form, but in place of d it has $d + 2\theta l + \theta^2 a$, which by properly determining θ can be made $= 0$. And then changing the original l, m, n , the equation is

$$\nabla = \begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & 0 \end{vmatrix} = 0.$$

Or, writing for shortness,

$$K = \begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix},$$

and denoting the minors hereof by (a, b, c, f, g, h) , then the equation is

$$\nabla = (a, b, c, f, g, h)(l, m, n)^2 = 0,$$

where the degree of K is 4, and the degrees of a, b, c, f, g, h are 4, 2, 2, 2, 3, 3 respectively, those of l, m, n being 0, 1, 1 respectively.

The nodes are, as before, the intersections of the quadric surfaces $B = 0, C = 0, F = 0$, viz. (d being now $= 0$) the values are

$$-B = cl^2 - 2gln + an^2,$$

$$-C = bl^2 - 2hlm + am^2,$$

$$F = fl^2 - glm - hln + amn.$$

But, according to a previous remark, the nodes lie also on the quartic surface $K = 0$; viz. this is a set of four planes intersecting in the line $x = 0, y = 0$.

Now, in general, any plane through the line $x = 0, y = 0$ meets the surface in this line twice and in a conic; if the plane is $y = \theta x$, we have

$$a, b, c, f, g, h = a', b'x^2, c'x^2, f'x^2, g'x, h'x,$$

where a', b', c', f', g', h' are functions of θ of the degrees (0, 2, 2, 2, 1, 1) respectively; and thence also

$$a, b, c, f, g, h = a'x^4, b'x^2, c'x^2, f'x^2, g'x^3, h'x^3,$$

where a', b', c', f', g', h' are functions of θ of the degrees 4, 2, 2, 2, 3, 3 respectively; the equation of the surface thus becomes $(a', b', c', f', g', h')(lx, m, n)^2 = 0$; viz. this is a quadric equation which, combined with the equation $y - \theta x = 0$, determines the

conic in question. But for each of the planes $K=0$, we have $(a', b', c', f', g', h' \sqrt{lx, m, n})^2$ a perfect square, or the conic a two-fold line; we have thus the 8 nodes lying in pairs on four lines, say the four "rays," in the four planes $K=0$ respectively; each of these rays meets the double line $x=0, y=0$ in a point; and we have thus on the double line 4 points, which are in fact pinch-points of the surface (as to this presently). It has just been stated that for the plane passing through the nodal line and a ray, the conic is a two-fold line (the ray twice) containing upon it a pair of nodes; more properly, the conic is the point-pair composed of the two nodes.

We can find through the nodes four different plane-pairs; in fact, forming the equation

$$-B + 2\lambda F - \lambda^2 C = 0,$$

this is

$$l^2(c + 2\lambda f + \lambda^2 b) - 2l(g + \lambda h)(n + \lambda m) + a(n + \lambda m)^2 = 0;$$

or, as this may also be written,

$$[a(n + \lambda m) - l(g + \lambda h)]^2 + l^2(b - 2\lambda f + \lambda^2 c) = 0,$$

where b, c, f and therefore also $b - 2\lambda f + \lambda^2 c$ are of the form $(x, y)^2$; say that we have $b - 2\lambda f + \lambda^2 c = (p, q, r \sqrt{x, y})^2$, where p, q, r are of course quadric functions of λ ; determining λ by the quartic equation $pr - q^2 = 0$, we have $b - 2\lambda f + \lambda^2 c$ a perfect square, $= (\alpha x + \beta y)^2$ suppose; and we have thus the plane-pair

$$[a(n + \lambda m) - l(g + \lambda h)]^2 - l^2(\alpha x + \beta y)^2 = 0$$

containing the eight nodes; viz. there are four such plane-pairs. The two planes of a plane-pair intersect in a line called an "axis"; that is, we have four axes each meeting the nodal line; and we have thus also through the nodal line and the four axes respectively four planes, which are "pinch-planes" of the quartic surface (as to this presently).

It has just been seen that the equation $B - 2\lambda F + C\lambda^2 = 0$ (where λ is arbitrary) is expressible in the form

$$[a(n + \lambda m) - l(g + \lambda h)]^2 + l^2(p, q, r \sqrt{x, y})^2 = 0,$$

viz. this is the equation of a quadric cone having its vertex on the nodal line at the point $x=0, y=0, an - lg + \lambda(am - lh) = 0$; this is, in fact, a cone *touching* the surface, as at once appears by writing the equation of the cone in the form

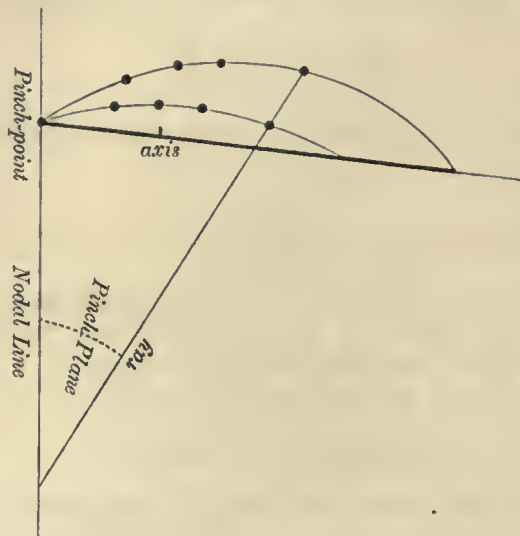
$$\frac{1}{C}\{BC - F^2 + (\lambda C - F)^2\} = 0,$$

that is,

$$\frac{1}{C}\{-l^2\nabla + (\lambda C - F)^2\} = 0;$$

we thus see that, taking for vertex any point whatever on the nodal line, there is a circumscribed quadric cone.

For each of the above-mentioned four values of λ , the quadric cone breaks up into a plane-pair; each plane of the plane-pair is thus a "trope" or plane touching the surface along a conic; viz. this is the conic passing through the intersection of the plane (or say of an axis) with the nodal line and through four nodes of the surface. We have thus 8 tropes, intersecting in pairs in the four axes (and the intersection of each axis with the nodal line being a pinch-point). Moreover, joining the nodes in pairs, we have four rays, each meeting the nodal line, the plane through it and the nodal line being a pinch-plane; this is illustrated in the figure.



As to the pinch-planes and pinch-points, remark first that a plane through the nodal line is in general a bitangent plane, its two points of contact being the points where the conic in such plane meets the nodal line. When the two points of contact come to coincide, the plane is a pinch-plane; viz. this happens when the plane passes through a ray, the conic being then the ray twice repeated. And secondly, at a point on the nodal line there are in general two tangent planes, viz. these are the tangent planes to the quadric cone belonging to such point; when the two tangent-planes come to coincide the point is a pinch-point, and this happens when the point is the intersection of the nodal line with an axis, for then (the quadric cone breaking up into the two tropes through the axis) the two tangent planes become the plane through the axis taken twice.

Each section through the nodal line is a conic, and the polar of the nodal line in regard to this conic is a point; the locus of this point (for different sections through the nodal line) is a right line which may be called simply the "polar." To prove this, considering the section by the plane $y = \theta x$, we have to find the pole of the line $x = 0$ in regard to the conic

$$(a', b', c', f', g', h' \chi l x, m, n)^2 = 0;$$

this is $lx : m : n = a' : h' : g'$, viz. if $g = g_0x + g_1y$, $h = h_0x + h_1y$, this is

$$lx : m : n = a : g_0 + g_1\theta : h_0 + h_1\theta,$$

or joining hereto the equation $y = \theta x$, we have

$$lx : ly : m : n = a : a\theta : g_0 + g_1\theta : h_0 + h_1\theta,$$

where l, a, g_0, g_1, h_0, h_1 are constants; m, n are linear functions of the coordinates (x, y, z, w) . The equations represent, it is clear, a right line which is the polar in question; and they may be written

$$\frac{lx}{a} = \frac{h_1m - g_1n}{h_1g_0 - h_0g_1}, \quad \frac{ly}{a} = -\frac{h_0m - g_0n}{h_1g_0 - h_0g_1}.$$

When the plane passes through a ray, the conic becomes, as was stated, the point-pair composed of the two nodes in such ray; the harmonic in regard to these two points of the intersection of the ray with the nodal line is thus a point on the polar: that is, the polar meets the ray; and the two nodes are situate harmonically in regard to the intersections of the ray with the nodal line and the polar respectively.

The polar may be arrived at in a different manner, viz. if instead of a plane through the nodal line we consider a point on the nodal line, this is the vertex of a circumscribed quadric cone; and taking the polar plane of the nodal line in regard to this cone, then considering the point as variable, the different polar planes all pass through a line which is the polar in question. And hence, taking for the point the intersection of the nodal line with an axis, it appears that the axis meets the polar; and, moreover, that the two tropes through the axis are harmonics in regard to the planes through the axis, and the polar and nodal line respectively.

Collecting the foregoing results, we have a quartic surface as follows:

We have two lines, a nodal line and a polar; meeting each of these, four lines called "rays" and four other lines called "axes." On each ray, harmonically in regard to its intersections with the nodal line and the polar, two nodes of the surface (in all 8 nodes): through each axis, harmonically in regard to the planes through it and the nodal line and the axis respectively, two tropes of the surface (in all 8 tropes). In each trope (or, what is the same thing, in its conic of contact) are 4 nodes; through each node (or, what is the same thing, touching its tangential quadricone) are 4 tropes; the relation of the nodes and tropes may be thus represented, viz. taking the pairs of nodes to be 1, 2; 3, 4; 5, 6; 7, 8; and those of tropes to be I, II; III, IV; V, VI; VII, VIII; then we have